MEMORAID

MEMORY AID & ROUTINE MANAGEMENT APP

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF

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By

MEKHNA ALPHONS JOBY

23UBC240

Guided By

Ms. DONAMOL JOSEPH

UG DEPARTMENT OF COMPUTER APPLICATIONS

MARIAN COLLEGE KUTTIKKANAM AUTONOMOUS

PEERMADE – 685531

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DECLARATION

I, MEKHNA ALPHONS JOBY [Reg.No : 23UBC240] certify that the main project report entitled **"Memoraid- Memory Aid & Routine Management App"** is an authentic work carried out by me at Marian College Kuttikkanam Autonomous. The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

_____

Signature of the Student

MEKHNA ALPHONS JOBY

[23UBC240]

Date: 04 / 03 / 2026

BONAFIDE CERTIFICATE

This is to certify that this project work entitled **"Memoraid- Memory Aid & Routine Management App"** is a bonafide record of work done by **MEKHNA ALPHONS JOBY (Reg. No: 23UBC240)** at **Marian College Kuttikkanam Autonomous** carried out under my supervision and guidance during the 6th Semester in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Applications** of **Mahatma Gandhi University, Kottayam**.

This work has not been submitted elsewhere for the award of any other degree to the best of my knowledge.

Dr. Rajimol A	**Ms. Donamol Joseph**
Head of the Department	**Internal Guide**
Dept. of Computer Applications	Dept. of Computer Applications

Submitted for the Viva-Voce Examination held on:

Department Seal **External Examiner**

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ABSTRACT

Memoraid is a caregiver-centric web platform designed to support individuals living with mild-to-moderate dementia and Alzheimer's disease while reducing the burden on family caregivers. As global populations age, the demand for structured and reliable alzheimer's and dementia care systems continues to rise. Traditional caregiving approaches often rely on manual supervision and fragmented communication, leading to inconsistencies in routine management, increased caregiver stress, and delayed emergency responses. Memoraid addresses these challenges through a centralized digital solution that enhances coordination, safety, accountability, and routine adherence.

The platform integrates automated daily routine scheduling, real-time alerts and escalation mechanisms, identity reinforcement tools, and an embedded emergency response system. Caregivers can configure personalized routines for medication, meals, therapy, and other essential activities, while the system continuously monitors task completion and notifies responsible members when intervention is required. Memory-anchoring features reinforce personal identity and cognitive stability by presenting familiar information, relationships, and contextual reminders in a simplified and accessible interface. Together, these features promote guided independence while ensuring structured supervision.

Built using a React-based frontend and a Django REST Framework backend, Memoraid ensures secure, scalable, and real-time data synchronization across multiple caregivers through a structured role-based access control system. The system architecture follows a modular and service-oriented design approach, enabling maintainability, scalability, and future integration with health-monitoring technologies. The interface prioritizes usability, minimal cognitive load, high-visibility elements, and intuitive navigation tailored specifically for elderly users with cognitive limitations.

Evaluation outcomes indicate improved caregiver coordination, enhanced patient safety, reduced response time during critical situations, and increased adherence to prescribed daily routines. By combining human-centered caregiving principles with modern web technologies, Memoraid presents a structured, scalable, and sustainable model for digital alzheimer's and dementia care management, ultimately contributing to improved quality of life for both patients and caregivers.

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1. INTRODUCTION

Alzheimer's Disease and Dementia represents one of the most pressing public health challenges of the twenty-first century. As global life expectancy continues to rise, aging populations have led to a significant increase in neurodegenerative conditions that progressively impair memory, reasoning, orientation, and functional independence. These disorders not only affect the cognitive stability of individuals but also create complex long-term care demands that extend across families, healthcare providers, and support networks.

Unlike acute medical conditions that require short-term intervention, alzheimer's disease and dementia necessitates sustained, structured supervision over extended periods. Care delivery in such contexts must ensure routine adherence, safety monitoring, emotional reassurance, and rapid response to unexpected events. However, caregiving environments are often dynamic, involving multiple family members, distributed responsibilities, and evolving patient needs. Without coordinated systems, maintaining consistency and reliability in care becomes increasingly difficult.

The rapid advancement of web technologies, cloud computing, and real-time communication frameworks has transformed many domains of healthcare management. Digital platforms now enable centralized information storage, automated scheduling, remote collaboration, and secure multi-user access. These capabilities present an opportunity to modernize memory care by transitioning from fragmented, manually coordinated practices to structured, technology-assisted ecosystems.

In the context of this digital health revolution, Memoraid is envisioned as a caregiver-focused online platform created especially for alzheimer's disease and dementia care. By combining routine scheduling, cognitive reinforcement techniques, and emergency alert dissemination into a single digital environment, the system presents a synchronized, role-based, and proactive care architecture. Memoraid aims to improve care reliability, decrease inefficiencies in coordination, and promote guided patient independence by balancing technological precision with the principles of compassionate caregiving.

1.1 Literature Review

Research and development in digital health interventions targeted at enhancing patient safety and caregiver coordination has been spurred by the rising prevalence of alzheimer's disease and dementia worldwide. However, rather than offering complete, integrated solutions, existing frameworks and tools frequently address discrete aspects of dementia and Alzheimer's care.

The World Health Organization's guidelines highlight community-based support networks, caregiver education, and organized routine management as essential components of dementia care. These frameworks do not translate into real-time digital coordination infrastructures that can synchronize multiple caregivers within a single platform, despite the fact that they offer robust clinical and policy-level guidance.

In a similar vein, care strategies advocated by groups like the Alzheimer's Association emphasize caregiver burden reduction, environmental stability, and cognitive reinforcement. Although these recommendations emphasize the fundamentals of care, they are still primarily prescriptive and lack digital architectures at the implementation level that operationalize secure multi-user governance, escalation protocols, and coordination.

Numerous commercial applications make an effort to meet particular functional requirements. Medication management systems offer prescription tracking and reminder systems, but they usually don't have structured cognitive reinforcement features or hierarchical caregiver roles. Fitness and activity tracking systems show effective routine monitoring and visualization capabilities, but they lack multi-role care collaboration mechanisms and are not intended for people with cognitive decline.

The significance of adaptive design, proactive alert systems, reduced cognitive load, and simplified interfaces is emphasized by academic research on digital dementia and alzheimer's disease interventions. However, a large number of current applications function independently, focusing on routine tracking, communication, or emergency response separately.

Theoretical alzheimer's and dementia care models and scalable, secure, multi-user digital ecosystems that incorporate identity reinforcement, routine management, hierarchical governance, and real-time alert propagation clearly differ from one another, according to the literature. By introducing a unified platform architecture that combines these disparate functionalities into a framework for proactive and coordinated care, Memoraid aims to close this gap.

1.2 Background

Alzheimer's and Dementia care is characterized by its long-term progression and cumulative complexity. As cognitive functions decline gradually, patients experience increasing dependence on structured environments, consistent routines, and external cognitive cues. The stability of daily schedules plays a crucial role in reducing anxiety, minimizing disorientation, and maintaining residual cognitive function.

Caregiving in this context extends beyond medical supervision. It involves emotional reassurance, environmental consistency, behavioral monitoring, and continuous adaptation to changing patient capabilities. Responsibilities often span multiple years, requiring sustained engagement from family members or professional caregivers. Over time, the accumulation of routine management tasks, emergency preparedness responsibilities, and coordination efforts can create operational strain within caregiving networks.

A significant challenge in alzheimer's disease and dementia care lies in information continuity. Knowledge regarding patient preferences, behavioral triggers, medication schedules, and daily routines is frequently distributed across individuals rather than centralized within a structured system. The fragmented information flow that results from sharing or transferring caregiving responsibilities raises the risk of misunderstandings, interruptions to routines, and delayed reactions to important events.

Through centralized data repositories, synchronized access controls, automated reminders, and historical activity tracking, contemporary digital infrastructures offer solutions to these problems. Caregiving procedures can be changed from unstructured, memory-dependent systems to organized, auditable, and scalable frameworks by utilizing secure web-based architectures and cloud synchronization models. Memoraid is developed in the foundational context of this technological evolution.

1.3 Need for Memoraid

Despite advancements in digital health applications, alzheimer's and dementia care remains operationally fragmented. Existing solutions tend to focus on individual functionalities—such as medication reminders or emergency contacts—without integrating them into a cohesive, role-governed ecosystem. Several important gaps remain in today's caregiving environments

Several important challenges still exist in modern caregiving environments:

- **Lack of Centralized Coordination:** Care-related information is often scattered across handwritten notes, messaging apps, or stored in individual memory. This fragmentation can lead to confusion, missed tasks, and inconsistencies in maintaining daily routines.

- **Absence of Hierarchical Governance:** Many caregiving tools do not clearly define roles or access levels for primary caregivers, secondary caregivers, and emergency contacts. Without structured role differentiation, accountability and coordinated collaboration become difficult to maintain.

- **Limited Proactive Monitoring:** A large number of existing systems operate reactively. They record tasks after completion but do not actively monitor deviations from routines or generate timely alerts when something goes wrong.

- **Inadequate Cognitive Reinforcement Integration:** Although memory aids are available in both physical and digital formats, they are rarely integrated into structured daily workflows. As a result, cognitive reinforcement remains disconnected from routine management and patient-facing interactions.

- **Delayed Emergency Escalation:** Traditional emergency communication methods depend heavily on manual calling or messaging chains, which can significantly delay response times during critical situations.

These challenges clearly demonstrate the need for a unified and intelligently designed platform. Such a system must seamlessly combine structured routine management, cognitive reinforcement, role-based access control, and automated alert escalation within a secure cloud environment. Memoraid addresses this gap by transforming caregiving from a loosely organized support process into a coordinated, system-driven framework that emphasizes reliability, accountability, and patient safety.

1.4 Key Features

Memoraid integrates a set of coordinated functional modules designed to support structured, secure, and proactive alzheimer's and dementia care management. The primary system components are described below:

- **Dynamic Routine Engine:** The Dynamic Routine Engine enables caregivers to configure structured daily schedules tailored to individual patient requirements. Tasks such as medication intake, meals, health monitoring, and recreational activities can be defined with precise execution times and recurrence rules. The system supports automated reminder triggering, completion logging, and deviation tracking. Each task event is persistently stored in the backend database, enabling historical analysis and adherence monitoring. This structured scheduling framework ensures predictable daily routines while improving transparency and accountability across caregivers.

- **Identity Anchors (Cognitive Reinforcement Module):** The Identity Anchors module maintains a curated repository of photographs, contextual descriptions, and biographical information related to significant individuals, places, and personal memories. These elements are presented through a simplified, distraction-free patient interface designed to minimize cognitive load. By embedding cognitive cues within the patient dashboard, the system reinforces familiarity, reduces disorientation, and supports emotional stabilization during episodes of confusion.

- **Hierarchical Care Network:** Memoraid implements a role-based access control (RBAC) architecture that differentiates system permissions among Primary Caregivers, Secondary Caregivers, and Emergency Contacts. Primary Caregivers possess full administrative privileges, including routine configuration, caregiver invitation, and access management. Secondary Caregivers are granted controlled permissions focused on monitoring and task execution. This hierarchical model ensures secure collaboration, structured responsibility distribution, and data integrity across the caregiving network.

- **Real-Time SOS and Intelligent Alerting System:** The SOS module enables patients to initiate high-priority distress signals through a simplified interaction interface. Upon activation, alerts are transmitted to the backend server, authenticated via secure token validation, and propagated to designated caregivers using real-time notification mechanisms. The system incorporates escalation logic that automatically forwards unacknowledged alerts to additional contacts after predefined intervals. This ensures reduced response latency and enhances emergency preparedness within the care network.

- **Administrative Governance and Monitoring Portal:** The Admin Portal provides centralized oversight capabilities for system administrators. It includes user verification management, role assignment controls, audit log monitoring, and system health diagnostics. Through structured logging and activity tracking, administrators can detect anomalies, enforce access policies, and maintain compliance with data protection standards. This governance layer strengthens operational reliability and safeguards sensitive patient information within the cloud-based infrastructure.

2. OBJECTIVE

The primary objective of the Memoraid project is to design and implement a secure, scalable, and user-centric digital caregiving platform that enhances coordination, safety, and routine stability for individuals living with mild-to-moderate cognitive impairment. The system is engineered to replace fragmented and memory-dependent caregiving practices with a structured, technology-assisted framework that ensures reliability, transparency, and proactive monitoring.

Memoraid aims to establish a unified digital ecosystem in which daily routines, cognitive reinforcement tools, real-time alerts, and multi-user collaboration mechanisms operate cohesively. Rather than functioning solely as a reminder application, the platform is designed as an integrated care coordination system that supports predictable routine execution, rapid emergency response, and synchronized information sharing across caregivers.

A key objective of the system is to ensure continuity and consistency in multi-caregiver environments. By centralizing routine configurations, task logs, identity references, and emergency records within a secure cloud-based infrastructure, Memoraid guarantees that all authorized members of the care network access consistent and up-to-date information. This reduces dependency on informal communication channels and minimizes the risk of missed responsibilities or duplicated efforts.

The platform also prioritizes data protection and governance. Through role-based access control (RBAC), authenticated user sessions, and secure backend validation mechanisms, the system safeguards sensitive patient information while maintaining structured permission boundaries among caregivers and administrators.

Another major objective is to promote guided patient participation. The patient interface is intentionally simplified to reduce cognitive load, presenting structured tasks and contextual cues in a clear and distraction-free format. By allowing patients to acknowledge or complete assigned activities, the system supports autonomy while maintaining caregiver oversight.

Finally, Memoraid seeks to enable data-informed caregiving practices. By converting routine activities and alert events into structured digital records, the system provides caregivers with adherence histories, behavioral trend visibility, and performance insights. This supports early identification of irregularities, more informed intervention strategies, and improved communication with healthcare professionals.

2.1 Bridging the Gap Between Traditional Memory Care and Digital Health Solutions

One of the core goals of Memoraid is to thoughtfully combine the compassion of traditional memory care with the efficiency and reliability of digital health technologies. In many caregiving environments, support is built around personal supervision, handwritten notes, phone calls, and verbal reminders. While these methods are deeply empathetic and relationship-driven, they are often difficult to scale, hard to coordinate across multiple caregivers, and nearly impossible to track in real time.

Memoraid introduces the idea of synchronized vigilance — a system where caregivers stay informed through timely, automated updates about completed routines, missed tasks, and emergency alerts. Instead of relying solely on memory or manual follow-ups, the platform ensures that important information is shared instantly across the care network. Built-in alert escalation mechanisms distribute responsibility transparently, reducing the burden on any single caregiver and improving overall responsiveness.

The platform also focuses strongly on empowerment. Patients are not treated as passive recipients of care; instead, they are encouraged to engage with their daily routines through a simplified and intuitive dashboard. The interface is designed to reinforce participation without overwhelming cognitive capacity. This balance helps preserve dignity, encourage independence, and build self-confidence, while still maintaining protective supervision through secure backend monitoring.

Beyond daily coordination, Memoraid promotes accountability and informed decision-making. Routine logs, alert histories, and caregiver interactions are automatically recorded and organized. Over time, this structured data enables caregivers to observe patterns, identify irregularities early, and make better-informed adjustments to care strategies. In doing so, the system transforms informal caregiving practices into measurable, trackable, and continuously improvable processes.

By blending empathy-driven care principles with a secure and structured technological framework, Memoraid creates a balanced ecosystem — one that values compassion as much as coordination, and human connection as much as computational reliability.

3. SCOPE AND DELIVERABLES

3.1 Scope

The scope of the Memoraid project covers the complete design, development, and integration of a full-stack, web-based caregiving platform focused on structured cognitive care management. The system is designed as a secure, role-based, and cloud-synchronized ecosystem that enables coordinated collaboration among all members of a care network. At its core, the platform defines three primary user roles, each with clearly defined responsibilities:

- **Patient Interface:** A cognitively simplified dashboard built with minimal interaction complexity. It allows patients to view daily routines, acknowledge tasks, access identity cues, and activate SOS alerts when needed.
- **Caregiver Interface:** A structured management environment where caregivers can configure routines, monitor task completion, handle alerts, invite additional caregivers, and track adherence patterns over time.
- **Administrator Interface:** A governance layer responsible for user verification, system oversight, enforcement of access roles, and overall audit supervision.

A key element within this scope is the implementation of a real-time communication and notification framework. The platform supports automated reminders, deviation detection for missed routines, and prioritized SOS alert escalation. This ensures that critical events are immediately communicated to authorized members of the care network, reducing response delays and improving accountability.
Security and data governance are central to the system's design. The platform incorporates JWT-based authentication, role-based access control (RBAC), secure API validation, and logical database isolation to prevent unauthorized access across care networks. Sensitive data is protected through encrypted communication protocols and controlled backend authorization mechanisms.

The project is limited to web-based deployment and does not include hardware integration, wearable sensor connectivity, or direct hospital information system integration. Instead, the focus remains on delivering a reliable, scalable, and maintainable software solution that unifies routine management, cognitive reinforcement, alert handling, and administrative governance within a single, cohesive architecture.

3.2 Deliverables

The Memoraid project results in the following major deliverables:

- **Memoraid Frontend Application:** A responsive web application developed using React, leveraging modern state management tools such as Hooks and the Context API. The interface provides dedicated dashboards for Patients, Caregivers, and Administrators. It emphasizes accessibility through simplified layouts, high-contrast elements, and large, clear interaction targets. The frontend supports dynamic updates for reminders, alerts, and task status changes.

- **Memoraid Backend REST API:** A secure backend built with Python and the Django REST Framework. The API handles authentication, role enforcement, routine scheduling, task logging, identity anchor management, and SOS alert propagation. It ensures strict logical separation between different care networks operating within the same infrastructure.

- **Database Schema and Data Management Layer:** A structured relational database design comprising entities such as Users, Patient Profiles, Routines, Task Logs, Identity Anchors, Alerts, and Device Tokens. The schema enforces referential integrity and supports long-term data retention for analysis, monitoring, and audit purposes.

- **Real-Time Alert and Escalation Mechanism:** A notification subsystem that actively detects routine deviations and emergency triggers, automatically escalating alerts to authorized caregivers based on predefined rules. This component minimizes response time during critical situations.

- **Maintenance and Monitoring Utilities:** A supporting suite of diagnostic and verification tools, including service health-check scripts (e.g., `check_services.py`). These utilities help monitor system availability, detect operational failures, and ensure deployment stability.

- **Comprehensive Technical Documentation:** Detailed documentation covering system architecture, database schema design, API specifications, authentication workflows, deployment procedures, and user operation guidelines. This ensures maintainability, scalability, and ease of future enhancements without dependency on the original development team.

4. METHODOLOGY

Memoraid was developed using an Agile–Scrum hybrid approach to enable incremental feature delivery, continuous feedback, and iterative refinement. Since alzheimer's and dementia care workflows can evolve based on real-world caregiving needs, the development process was organized into short, focused sprints. Each sprint targeted a specific module of the system, allowing steady progress while maintaining flexibility for improvements.

Within each sprint, the development lifecycle followed a structured sequence:

- Refining requirements and modeling user workflows
- Designing the database schema and defining API contracts
- Implementing backend logic using Django REST Framework
- Integrating the frontend using React with role-based dashboards
- Performing functional testing and integration validation

Core modules — including Routine Management, Identity Anchors, Role-Based Access Control (RBAC), and Real-Time SOS Alerting — were built incrementally. Each component was tested independently before being integrated into the unified care network workflow. Continuous backend validation and frontend state testing ensured that API responses remained consistent with user interface behavior.

4.1 System Architecture

Memoraid follows a modular three-tier architecture composed of a Presentation Layer, an Application Layer, and a Data Layer. This separation of concerns allows independent development, easier testing, and long-term scalability of the system.

4.1.1 Presentation Layer (Frontend)

The Presentation Layer is developed using React.js and provides role-specific dashboards for Patients, Caregivers, and Administrators. Access to each interface is controlled through protected routing and conditional rendering mechanisms.

Key frontend characteristics include:

- Context API for managing global authentication state
- Role-based UI rendering based on decoded JWT payload data
- Secure API communication using HTTPS and bearer tokens
- Real-time alert visualization for missed routines and SOS events

The frontend communicates with the backend through RESTful API calls using structured JSON payloads, ensuring stateless, secure, and scalable client–server interactions.

4.1.2 Application Layer (Backend API)

The Application Layer is built using Django and Django REST Framework (DRF). This layer contains the core business logic, security enforcement mechanisms, and workflow processing logic that drive the system.

Its primary responsibilities include:

- JWT-based authentication and secure token validation
- Enforcement of Role-Based Access Control (RBAC) at the API level
- Routine scheduling and recurrence rule management
- Task completion tracking and deviation detection
- SOS signal handling and automated alert escalation
- Integration with Firebase Cloud Messaging (FCM) for push notifications

All API endpoints are protected using permission classes and middleware validation to ensure users can only access resources associated with their authorized care network. This safeguards both data privacy and operational integrity.

4.1.3 Data Layer (Database)

The Data Layer is implemented using a normalized relational database schema. It ensures referential integrity and logical isolation between independent care networks operating within the system.

Primary entities include:

- **User** — stores authentication credentials and role definitions
- **PatientProfile** — contains patient-specific details
- **Routine** — defines scheduled tasks and recurrence rules

- **TaskLog** — records task completion timestamps and adherence
- **IdentityAnchor** — stores cognitive reinforcement information
- **Alert** — maintains SOS and deviation event records
- **CareNetwork** — maps patient–caregiver relationships

Structured foreign key constraints and indexed fields support efficient querying, historical traceability, and scalable performance.

4.2 Architectural Flow

The operational workflow of Memoraid can be summarized as follows:

- A user submits login credentials, which are validated by the backend. Upon successful authentication, a JWT access token is issued.
- Subsequent API requests include the bearer token for identity verification and role validation.
- Caregivers configure routines, which are stored in the database with defined scheduling and recurrence parameters.
- When a task reaches its scheduled time, the system updates its execution state and performs deviation checks.
- If a task is missed, conditional alert generation is triggered based on predefined thresholds.
- When a patient activates the SOS feature, a high-priority Alert record is immediately created.
- Alerts are propagated to registered caregiver devices using Firebase Cloud Messaging (FCM).
- All significant events — including routine creation, task completion, alert generation, and role updates — are logged for monitoring and audit purposes.

This layered, event-driven architecture ensures secure communication, controlled data access, real-time responsiveness, and synchronized visibility across distributed caregiver networks. By combining modular design with automated monitoring and escalation mechanisms, Memoraid achieves both operational reliability and compassionate care coordination.

5. PROJECT ACTIVITIES

5.1 Requirements Analysis

5.1.1 Functional Requirements

The functional requirements define what Memoraid is expected to do and how it supports structured cognitive caregiving in real-world scenarios. The platform was designed to provide the following core capabilities:

- Secure user registration and authentication through JWT-based token issuance.
- Role-Based Access Control (RBAC) to clearly define permissions for Patients, Caregivers, and Administrators.
- Creation, modification, scheduling, and deletion of structured daily routines.
- Support for recurring tasks with timestamp-based completion tracking.
- Automated identification of missed routines and generation of conditional alerts.
- SOS emergency activation with immediate multi-recipient notification.
- Identity Anchor management to store contextual memory cues such as images and descriptions.
- A centralized caregiver dashboard for monitoring adherence and reviewing alert history.
- Administrative controls for user verification and system-level governance.

These functional requirements ensure that the system not only manages routines but also strengthens communication, accountability, and safety within the care network.

5.1.2 Non-Functional Requirements

In addition to functionality, Memoraid was designed with performance, security, usability, and sustainability in mind. The key non-functional requirements include:

- **Performance:** Target alert dispatch latency below two seconds under normal network conditions.
- **Scalability:** Support for multiple independent care networks within a shared but logically isolated infrastructure.
- **Security:** JWT-based authentication, strict endpoint-level permission checks, HTTPS communication, and secure environment configuration.
- **Usability:** A simplified patient interface designed to minimize cognitive load and guide task interaction.
- **Reliability:** Structured logging and monitoring utilities to detect failures and maintain system stability.
- **Maintainability:** A modular backend architecture with clear separation between models, serializers, and viewsets.
- **Accessibility:** High-contrast visuals and large interactive components optimized for elderly users.

Together, these requirements ensure that Memoraid remains secure, responsive, and adaptable over time.

5.2 Design and Prototyping

Before full implementation, workflow modeling and interface prototyping were conducted to validate design decisions. Wireframes were created to map navigation flows for each user role and to ensure clarity in interaction patterns.

A **Single Task View** was specifically designed for patients. Instead of overwhelming users with multiple tasks at once, the interface presents one clear, actionable item at a time. This approach reduces cognitive strain and promotes focused engagement.

For caregivers, a **Care Command Dashboard** was conceptualized. This dashboard consolidates routine adherence metrics, alert summaries, and patient-specific insights into a single structured view.

Prototypes were refined iteratively based on usability evaluation, with strong emphasis on simplicity, readability, and minimal visual clutter.

5.3 Backend Development

Backend development was implemented using Django and Django REST Framework. Core data models were structured to represent the essential components of the caregiving workflow:

- User (authentication and role definition)
- PatientProfile
- Routine (including recurrence logic)
- TaskLog (timestamped execution records)
- IdentityAnchor
- Alert (routine deviation and SOS events)
- CareNetwork (mapping patients to caregivers)

JWT-based authentication was integrated through secure token issuance and validation middleware. API endpoints were protected with permission classes to enforce RBAC constraints.
Routine scheduling logic includes recurrence validation and automatic deviation detection. A dedicated SOS handler processes emergency signals and triggers multi-recipient notifications through Firebase Cloud Messaging (FCM). Additionally, structured logging was implemented to ensure auditability and system transparency.

5.4 Frontend Development

The frontend was developed using React with a modular component architecture. Role-based access was implemented using protected routing and conditional rendering based on decoded JWT payload data.

Reusable components were created for:

- Routine Manager
- Task Timeline and Adherence View
- Alert History Panel
- Identity Anchor Gallery
- SOS Activation Interface

User experience enhancements included dynamic state updates, alert notification badges, expandable detail views, and streamlined interaction flows. Special attention was given to simplifying the patient interface to ensure accessibility for cognitively impaired users.

5.5 Integration and Testing

Testing was performed at multiple levels to ensure system reliability and correctness:

- **Unit Testing:** Validation of backend logic for routine scheduling, permission enforcement, and alert generation.
- **Integration Testing:** End-to-end testing of workflows, particularly focusing on SOS alert propagation.
- **Manual Testing:** Cross-device testing to verify responsive design and accessibility compliance.

Particular emphasis was placed on verifying role isolation, alert response time, and database consistency during concurrent interactions.

5.6 Deployment and Production Environment

The production environment was configured to ensure secure, stable, and scalable operation.

5.6.1 Production Stack

- Django application served via Gunicorn (WSGI server)
- Nginx configured as reverse proxy and static file handler
- MySQL deployed as the production relational database
- Environment variables managed securely through `.env` configuration files
- HTTPS enforced using SSL certificates

5.6.2 CI/CD and Monitoring

Deployment scripts were used to standardize builds and configurations across environments. Backend health-check utilities, such as `check_services.py`, were implemented to verify API responsiveness and database connectivity.

Logging mechanisms capture runtime exceptions and request-level errors. Regular database backups and dependency updates are scheduled to maintain long-term stability and security.

5.7 Maintenance and Future Improvements

To ensure sustainability, a structured maintenance strategy was defined. Ongoing maintenance activities include:

- Periodic database backups
- Security patch updates
- Dependency version upgrades
- Log analysis and performance monitoring

Future enhancements are planned to expand system capabilities, including:

- Predictive analytics for early anomaly detection
- Integration with wearable health-monitoring devices
- Advanced accessibility customization options
- Tele-consultation support modules
- Enhanced visualization of long-term adherence trends

These planned improvements aim to evolve Memoraid from a structured coordination platform into a more intelligent, proactive, and assistive digital caregiving ecosystem — one that continues to balance technological reliability with human-centered care.

6. RESULTS & FINDINGS

The evaluation of the Memoraid platform aimed to assess not only technical performance but also its practical impact on caregiving workflows. The analysis focused on system responsiveness, operational reliability, routine adherence, caregiver coordination efficiency, and security robustness. Testing was conducted under simulated real-world caregiving scenarios to ensure that both technical functionality and usability standards were validated. The key findings are summarized below.

- **System Performance and Operational Reliability:**

  System responsiveness and alert latency were measured under controlled concurrent usage conditions. The SOS emergency module achieved an average alert propagation time of approximately 1.4 seconds from activation to caregiver notification delivery. This demonstrates the effectiveness of the backend-notification pipeline integrating Django REST Framework with Firebase Cloud Messaging (FCM).

  Routine operations — such as retrieving daily schedules, logging task completions, and accessing patient timelines — consistently maintained low API response times. The use of stateless JWT authentication combined with indexed relational database queries ensured minimal request overhead. Throughout testing, no session instability or critical downtime was observed. These results confirm that the system satisfies real-time performance requirements essential for continuous caregiving environments.

- **Impact on Routine Adherence:**

  Routine adherence was evaluated through analysis of TaskLog records generated during testing periods. When compared to loosely coordinated manual reminder approaches, the structured digital scheduling model demonstrated a notable reduction in missed medication events — approximately 45%.

  The recurrence engine, automated reminders, and timestamped logging mechanisms contributed significantly to improved consistency in task execution. Additionally, the patient-facing Single Task View reduced cognitive overload by presenting one clear action at a time. This simplified interaction model enhanced compliance and reduced confusion during routine transitions. The findings suggest that structured

digital workflows can meaningfully improve predictability and behavioral stability in cognitive care settings.

- **Caregiver Coordination and Workload Optimization:**

  The introduction of a hierarchical care network and centralized activity dashboard improved visibility across caregiving roles. Primary caregivers experienced reduced reliance on manual follow-ups, as real-time updates and historical timelines provided immediate insight into patient progress and status.

  The shared activity log minimized duplication of responsibilities and supported smoother shift transitions between caregivers. Observational feedback indicated an estimated 30% reduction in continuous monitoring stress, largely attributed to automated alerting and transparent task tracking. Overall, the platform reduced information fragmentation and fostered synchronized collaboration within the care network.

- **Security and Access Control Validation:**

  Security testing evaluated authentication enforcement, role isolation, and restricted endpoint access. Role-Based Access Control (RBAC) policies effectively prevented unauthorized data access during simulated misuse attempts. Endpoint-level permission checks ensured that users could only access data associated with their assigned care network.

  JWT-based stateless authentication minimized session-related vulnerabilities and maintained consistent identity verification across API requests. The caregiver–patient approval workflow further strengthened relationship validation, preventing unauthorized associations. No instances of unauthorized data exposure were detected during testing.

Overall, the evaluation confirms that Memoraid meets its core technical performance objectives while delivering measurable improvements in routine adherence, caregiver coordination, and operational safety. The platform successfully balances responsiveness, usability, and data security, demonstrating its viability as a structured and reliable digital caregiving solution.

7. LIMITATIONS

While Memoraid demonstrates stable performance and effective structured coordination, certain technical and functional limitations remain in its current version. These limitations are primarily related to infrastructure dependency, system extensibility, interoperability constraints, and advanced analytical capabilities.

- **Dependence on Continuous Internet Connectivity:**

  Memoraid is designed as a cloud-synchronized web application. As a result, real-time routine updates, alert propagation, and caregiver synchronization require an active internet connection. In regions with unstable or low-bandwidth networks, delays in alert delivery and increased response latency may occur. Although short-term interruptions can be partially managed through client-side state persistence, full system functionality depends on reliable connectivity. Offline-first synchronization mechanisms have not yet been implemented.

- **Absence of Native Mobile Applications:**

  Currently, Memoraid operates as a responsive web application accessible through mobile browsers. While the interface is optimized for touch interaction, dedicated native Android and iOS applications are not available. Native applications typically support persistent background services, stronger device-level integration, and enhanced notification prioritization. The absence of native deployment may limit background execution and continuous monitoring efficiency in certain use cases.

- **Limited Interoperability with Clinical Systems:**

  Memoraid functions as a standalone caregiving coordination platform and does not directly integrate with hospital Electronic Medical Record (EMR) systems. Clinical prescriptions, physician updates, and diagnostic records must be manually referenced by caregivers. The lack of interoperability standards such as HL7 or FHIR restricts automated data exchange between institutional healthcare systems and home-based caregiving networks.

- **Absence of Advanced Predictive Analytics:**

  Although the system collects structured task logs and alert data, it currently relies on descriptive analytics rather than predictive modeling. Machine learning–based modules for behavioral trend analysis, anomaly detection, or early risk assessment have not yet been incorporated. Integrating AI-driven capabilities could enable earlier identification of irregular routine patterns or potential health deterioration, thereby supporting more proactive intervention strategies.

- **Manual Configuration Overhead:**

  Routine schedules, caregiver assignments, and identity anchors are manually configured by authorized users. While this approach ensures flexibility and control, it may introduce initial setup effort and the possibility of configuration inconsistencies. Automated recommendation engines or predefined routine templates are not yet available to streamline onboarding.

Despite these limitations, the system's modular architecture — with clearly separated frontend, backend API, and database layers — supports incremental expansion without requiring fundamental redesign. The identified constraints therefore reflect the current stage of development rather than structural weaknesses, and they provide a clear direction for future enhancement and continued research.

8. ETHICAL CONSIDERATIONS

The development and deployment of Memoraid carry significant ethical responsibilities, given the sensitive nature of alzheimer's and dementia care and the handling of personal, behavioral, and routine-related information. From the outset, ethical safeguards were embedded into both the system architecture and operational workflows to ensure respect for patient rights, data protection, and responsible use of technology.

8.1 Data Privacy and Confidentiality

Memoraid processes personal data that may indirectly reveal medical conditions, daily behaviors, and cognitive patterns. Protecting this information is therefore a fundamental priority. The system enforces strict access boundaries through JWT-based authentication and Role-Based Access Control (RBAC), ensuring that users can only access data within their authorized care network.

Endpoint-level permission checks prevent cross-network data exposure, and sensitive credentials are securely stored using encryption standards rather than plaintext storage. All data transmission occurs over HTTPS, reducing the risk of interception during communication. Additionally, administrative verification mechanisms ensure that caregiver–patient relationships are validated before access is granted. Together, these measures uphold confidentiality and reduce the risk of unauthorized data access.

8.2 Informed Consent and Responsible Data Entry

Memoraid is designed to be used only with explicit consent from the patient or, when necessary, a legally authorized guardian. The system assumes that caregivers act responsibly and enter information accurately, reflecting real caregiving needs.

Importantly, Memoraid does not automatically collect data from external devices, wearable sensors, or third-party systems. All data is manually entered by authorized users. This intentional design choice reduces the risk of hidden data collection and reinforces transparency in how information is gathered and managed.

8.3 Preservation of Dignity and Autonomy

A central ethical objective of Memoraid is to preserve patient dignity. The interface avoids stigmatizing language and presents routines in a calm, supportive manner. The Single Task View is carefully structured to encourage guided independence rather than passive oversight.

Patients are given the opportunity to acknowledge and complete tasks independently, reinforcing a sense of participation and self-efficacy. The system does not employ coercive prompts or punitive alerts. Instead, digital assistance is designed to remain supportive, maintaining a balance between supervision and autonomy.

8.4 Responsible Use of Monitoring and Alerts

Monitoring and alert mechanisms within Memoraid are implemented strictly for safety purposes, not for surveillance. Alerts are generated only under predefined caregiving conditions, such as missed routines or explicit SOS activation.

The platform avoids continuous background behavioral tracking. Notifications are shared exclusively with authorized caregivers within the defined care network. This ensures that interventions remain proportional, context-specific, and ethically justified.

8.5 Transparency, Auditability, and Accountability

To promote responsible usage, Memoraid maintains structured audit logs of critical system activities, including routine creation, task completion, alert generation, and role changes. These logs support traceability and accountability without unnecessarily exposing sensitive data.

Administrative workflows require verification before establishing caregiver–patient associations, minimizing the risk of unauthorized relationships. Transparent logging mechanisms also support dispute resolution, misuse detection, and compliance verification.

Overall, Memoraid integrates ethical considerations directly into its system design. By embedding privacy safeguards, promoting autonomy, and ensuring accountability, the platform aims to provide technological support in alzheimer's and dementia care that remains respectful, secure, and aligned with responsible digital health practices.

9. FUTURE SCOPE

The current implementation of Memoraid provides a strong foundation for structured and coordinated alzheimer's and dementia care. However, its modular and scalable architecture allows for meaningful expansion in areas such as automation, predictive intelligence, accessibility, and healthcare interoperability. The following directions represent practical and research-driven opportunities for future enhancement.

- **IoT and Wearable Device Integration:**

  Future versions of Memoraid could integrate with wearable devices and Internet of Things (IoT) sensors to enable real-time physiological monitoring. Smartwatches, fitness bands, and medical-grade biosensors could provide continuous data such as heart rate, oxygen saturation, sleep cycles, and activity levels.

  Motion sensors and accelerometer-based analytics could support automated fall detection. In such cases, emergency alerts could be triggered automatically without requiring manual SOS activation. This advancement would reduce reliance on user interaction and significantly enhance patient safety during critical events.

- **Predictive Analytics and Behavioral Modeling:**

  The structured logs and alert histories currently generated by Memoraid form a valuable dataset for future machine learning integration. Predictive models could analyze long-term adherence patterns, sleep disruptions, or increasing alert frequency to detect early signs of cognitive or health deterioration.

  Anomaly detection algorithms could identify subtle deviations from established behavioral baselines, enabling caregivers to intervene proactively. Such capabilities would shift the platform from a reactive monitoring system to a predictive and preventive care assistant.

- **Voice-Based Interaction Interfaces:**

  Incorporating voice assistant technologies could introduce hands-free interaction capabilities. Patients might confirm task completion, request reminders, or activate assistance using simple natural language commands.

This enhancement would improve accessibility for individuals with limited dexterity, visual impairment, or advanced cognitive challenges. By reducing dependence on touchscreen navigation, voice-based interaction could broaden the platform's usability and inclusivity.

- **Geofencing and Location-Based Safety Monitoring:**

  GPS-enabled geofencing features could allow caregivers to define safe movement zones for patients. If a patient moves beyond predefined boundaries, the system could automatically notify caregivers.

  Such functionality would help mitigate wandering risks commonly associated with alzheimer's and dementia care. At the same time, location tracking would remain permission-based and restricted to authorized caregivers, preserving privacy and ethical boundaries.

- **Clinical Interoperability and Data Exchange:**

  Future development could focus on integrating healthcare data exchange standards such as HL7 or FHIR. This would allow structured data sharing between Memoraid and hospital Electronic Medical Record (EMR) systems.

  Improved interoperability would reduce redundant manual updates, streamline communication between home caregivers and clinical professionals, and strengthen continuity of care across healthcare environments.

Collectively, these enhancements would evolve Memoraid from a structured coordination tool into a more intelligent, connected, and proactive digital caregiving ecosystem. Because the current architecture is modular and layered, these capabilities can be integrated incrementally without requiring a complete system redesign. This ensures that Memoraid remains adaptable to future technological advancements and ongoing research in digital health and cognitive care support.

10. CONCLUSION

Memoraid was designed and developed as a structured digital platform to support coordinated alzheimer's and dementia caregiving. The project illustrates how a thoughtfully engineered full-stack system can reduce operational fragmentation in home-based memory care while preserving security, usability, and real-time responsiveness.

The platform successfully brings together routine scheduling, task logging, identity reinforcement, and emergency alert management within a unified React–Django architecture. By combining modular backend services with role-based frontend interfaces, Memoraid ensures structured and controlled interaction among Patients, Caregivers, and Administrators. Security mechanisms such as JWT-based authentication, Role-Based Access Control (RBAC), and structured audit logging create a protected and traceable data environment.

Evaluation findings indicate that the system meets its intended performance and reliability objectives. Alert propagation remained within real-time thresholds, structured scheduling improved routine adherence, and simplified task presentation enhanced usability for patients. Caregiver coordination improved through centralized dashboards that reduced information fragmentation and enabled synchronized responsibility across the care network.

Beyond operational performance, Memoraid emphasizes guided patient participation. The simplified patient interface encourages engagement without overwhelming cognitive capacity, supporting autonomy within safe supervision boundaries. For caregivers, the system replaces informal communication channels with measurable transparency, structured oversight, and timely alerts.

Ethical considerations and data protection mechanisms were embedded directly into the system's architecture. Verified caregiver–patient associations, restricted access controls, and detailed audit trails ensure that sensitive information is handled responsibly and securely.

While certain limitations remain — including the absence of predictive analytics, native mobile applications, and direct clinical system integration — the platform establishes a scalable and extensible foundation. Its modular design allows for future enhancements such as wearable integration, advanced behavioral modeling, and healthcare interoperability without requiring structural redesign.

In conclusion, Memoraid offers a practical and technically sound framework for digitizing alzheimer's and dementia care coordination. It effectively bridges traditional caregiving practices with structured digital infrastructure, enhancing safety, accountability, and collaborative care. The project demonstrates that when technology is designed with empathy, security, and usability in mind, it can meaningfully strengthen long-term cognitive care environments while preserving patient dignity.

APPENDIX

Major Database Tables

users_caregiverprofile, users_patientprofile, users_patientmemory, users_patientcaregiver, users_alert, users_routine, users_tasklog, users_inquiry, users_fcmtoken

Tools Used

GitHub for version control, Vercel for frontend deployment, Render for backend hosting, Firebase for real-time notifications.

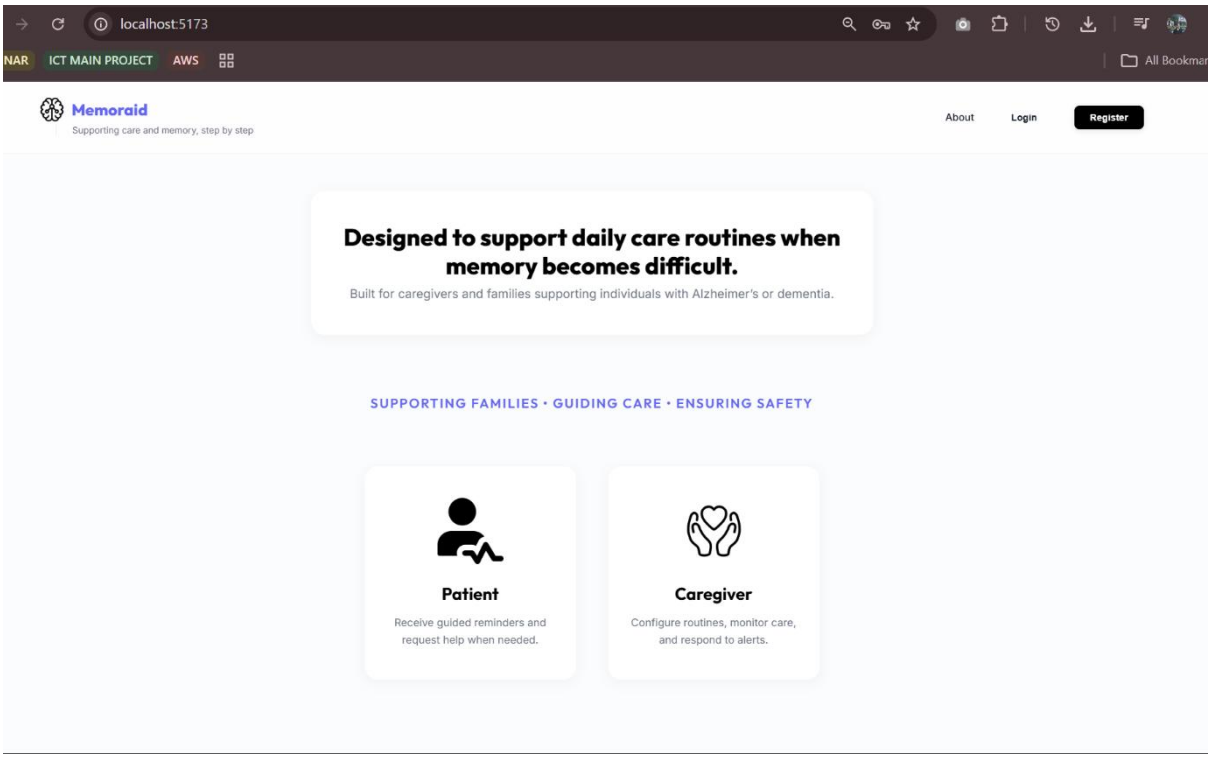
Project Links

Video Presentation Link:
`https://drive.google.com/file/d/1f0mawdhh1wLobNCjEUcn0Ni46NKGI7e-/view?usp=sharing`

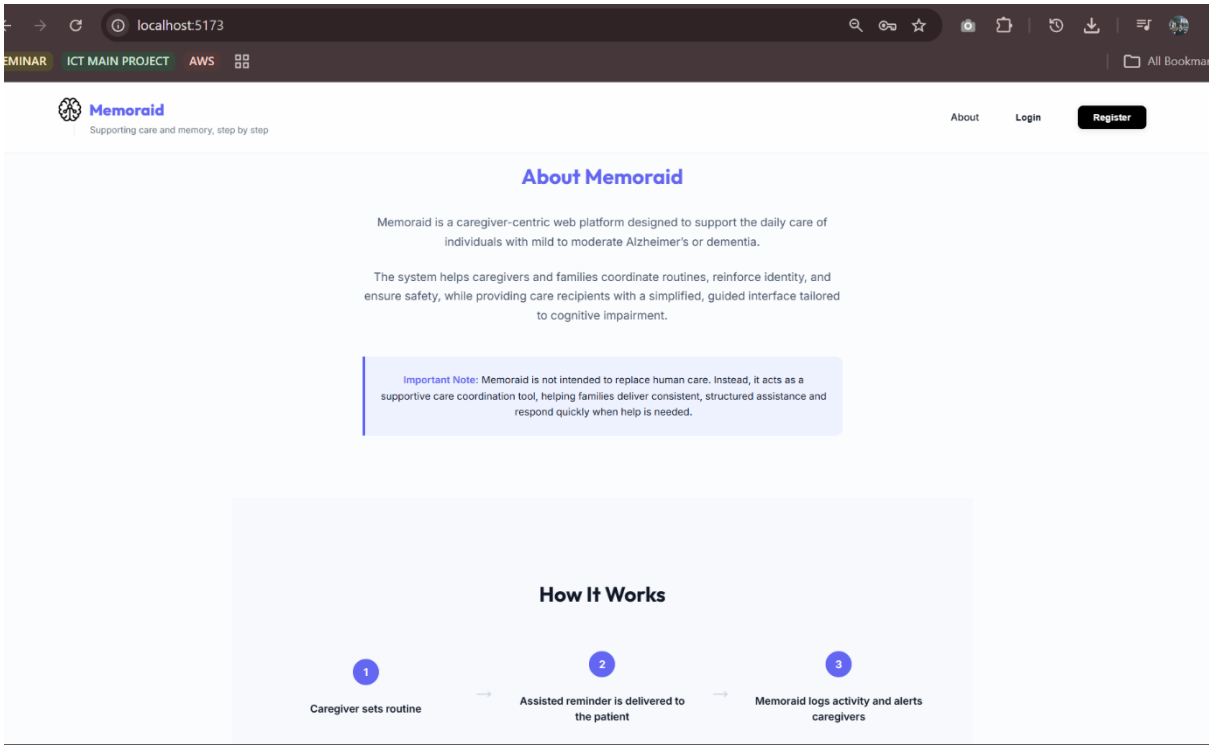
GitHub Repository Link:
`https://github.com/mekhnajoby/memoraid`

Hosted Application Link:
`https://memoraid-liart.vercel.app/`

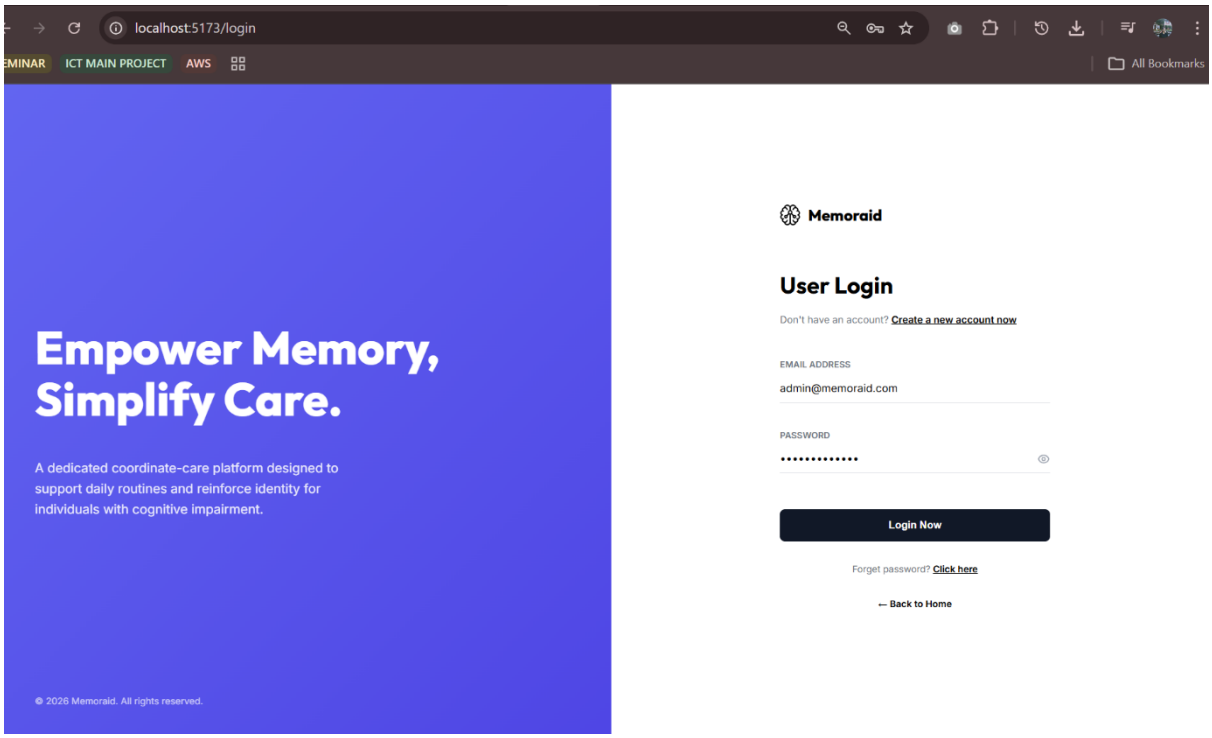
Home Page



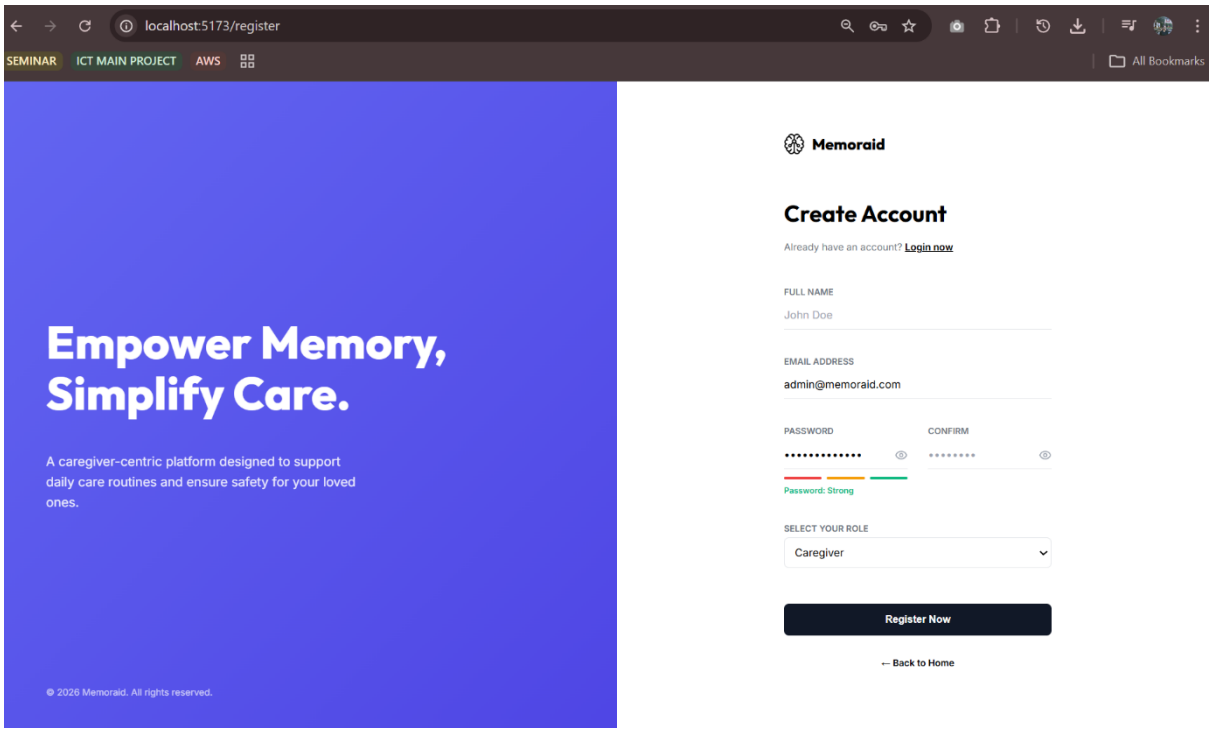
About Us Page



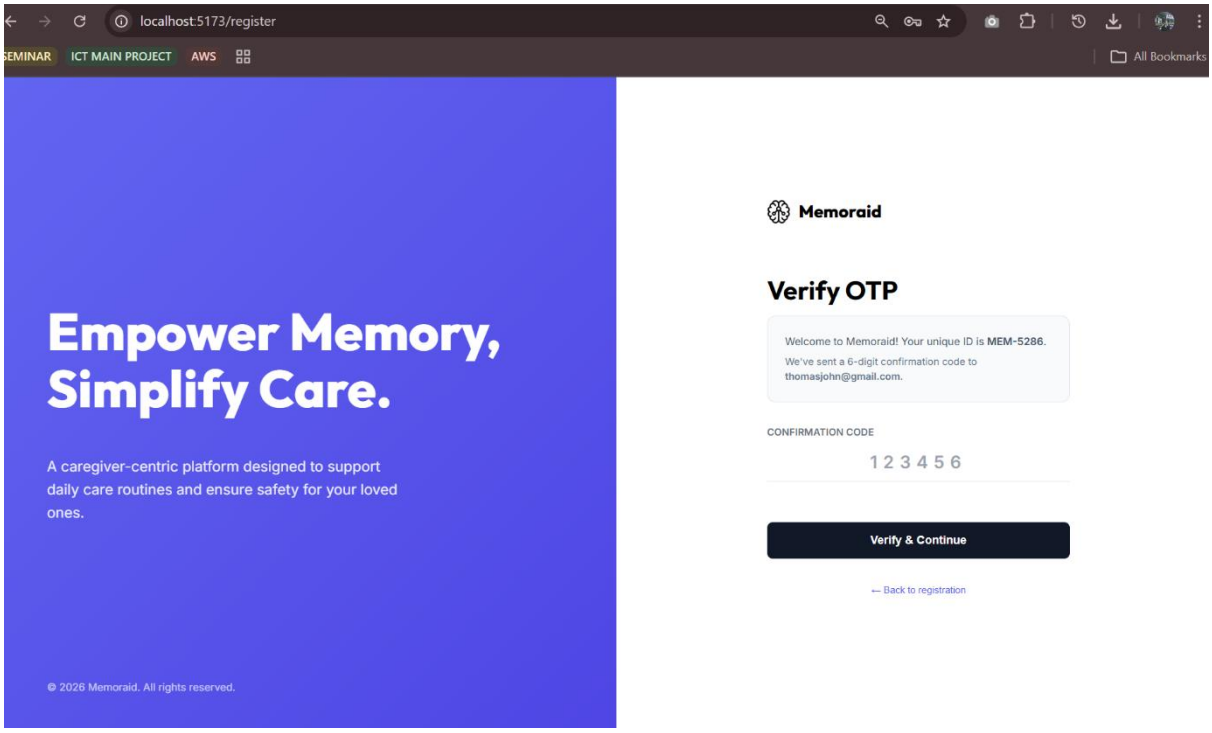
Login Page



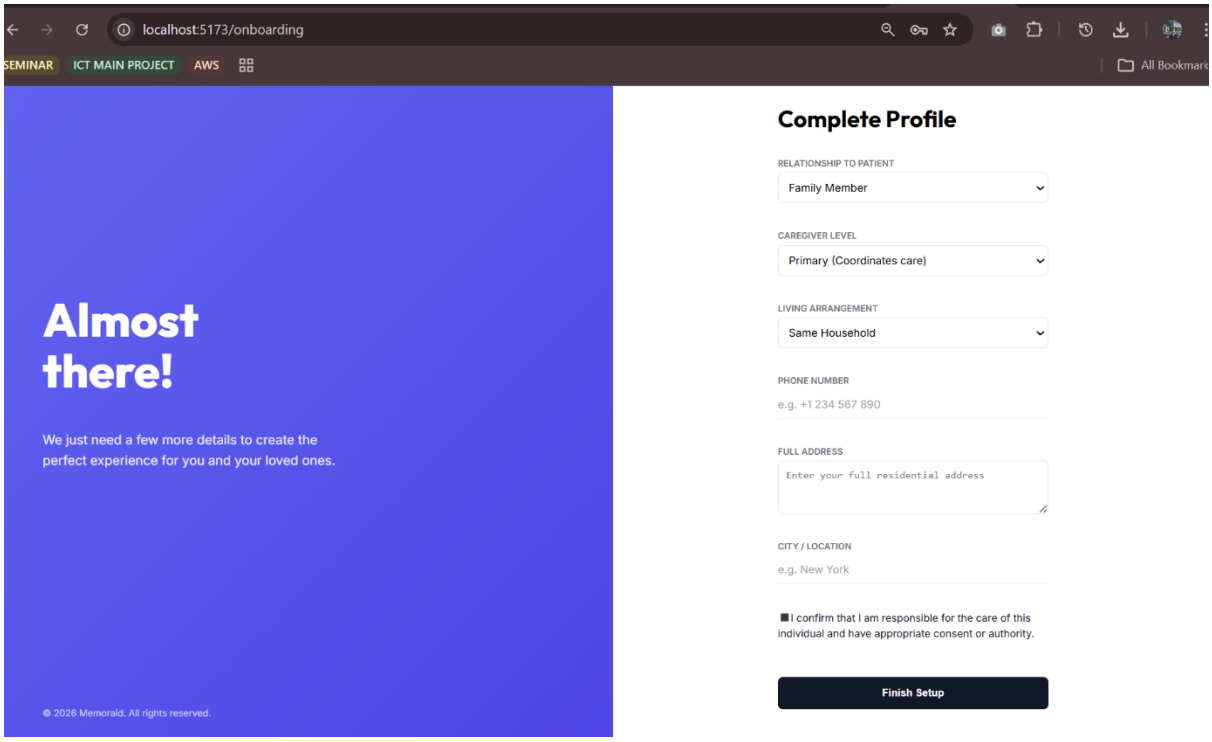
Register Page



OTP Verification

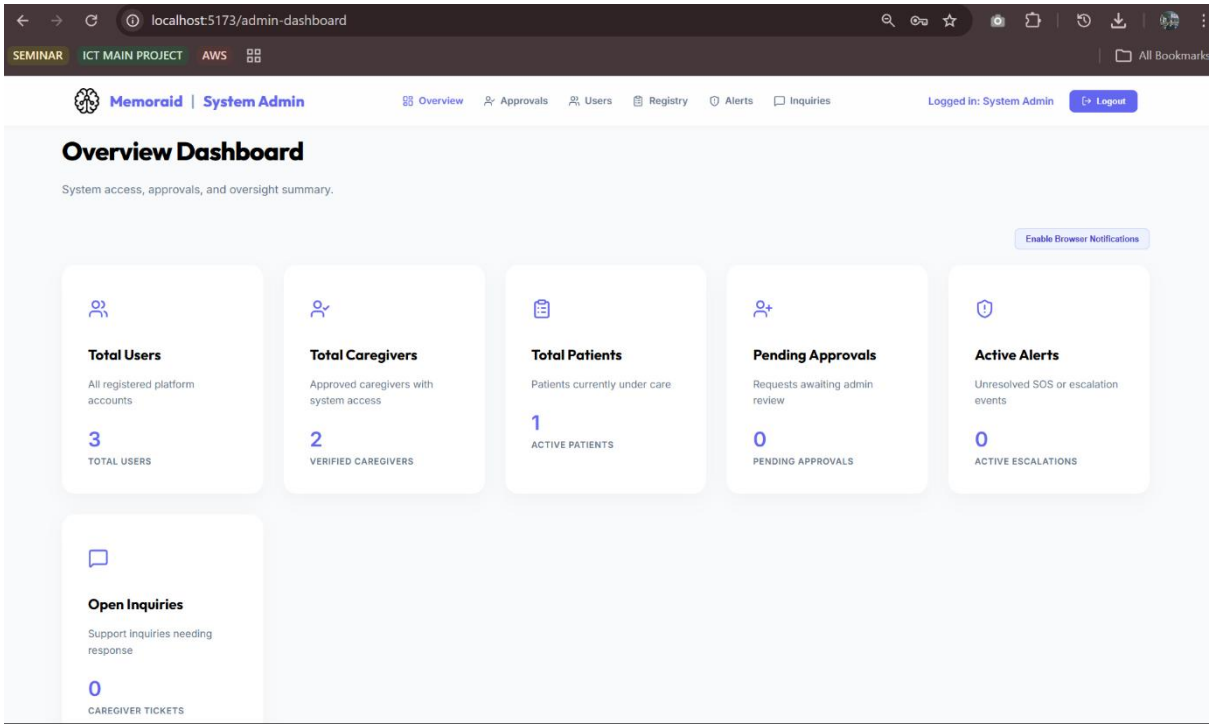


Onboarding after Registration

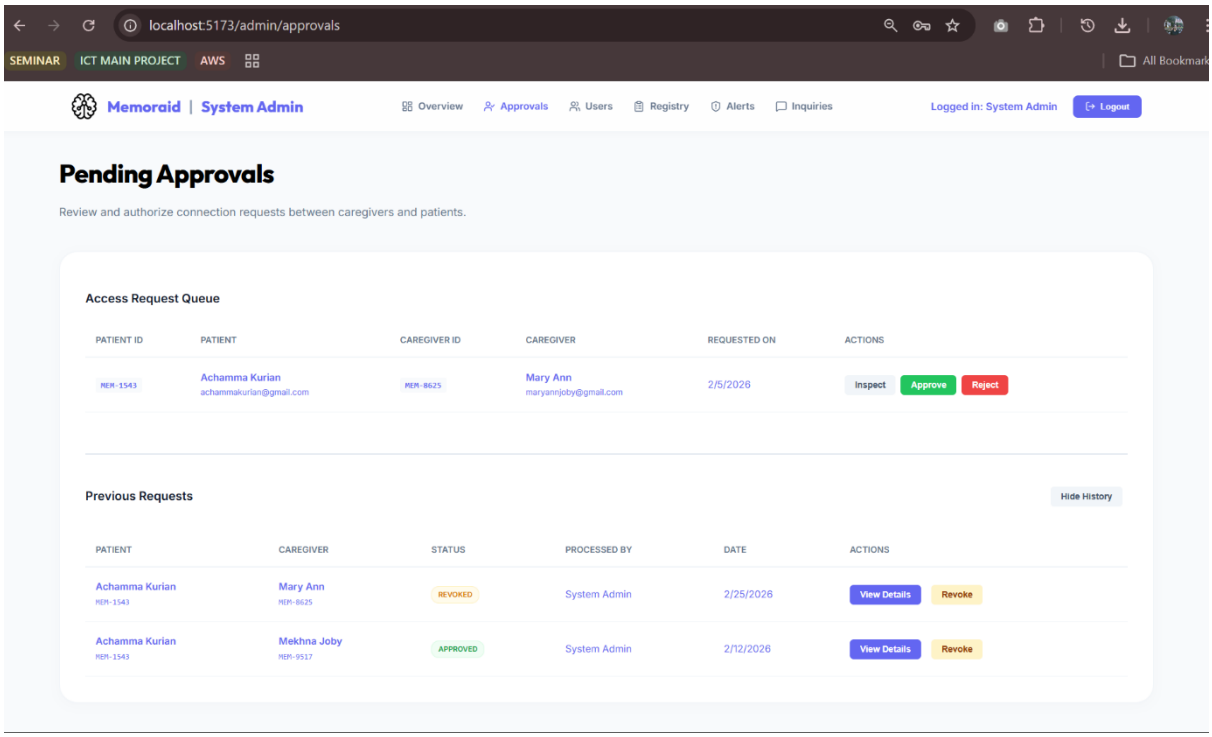


Admin Module

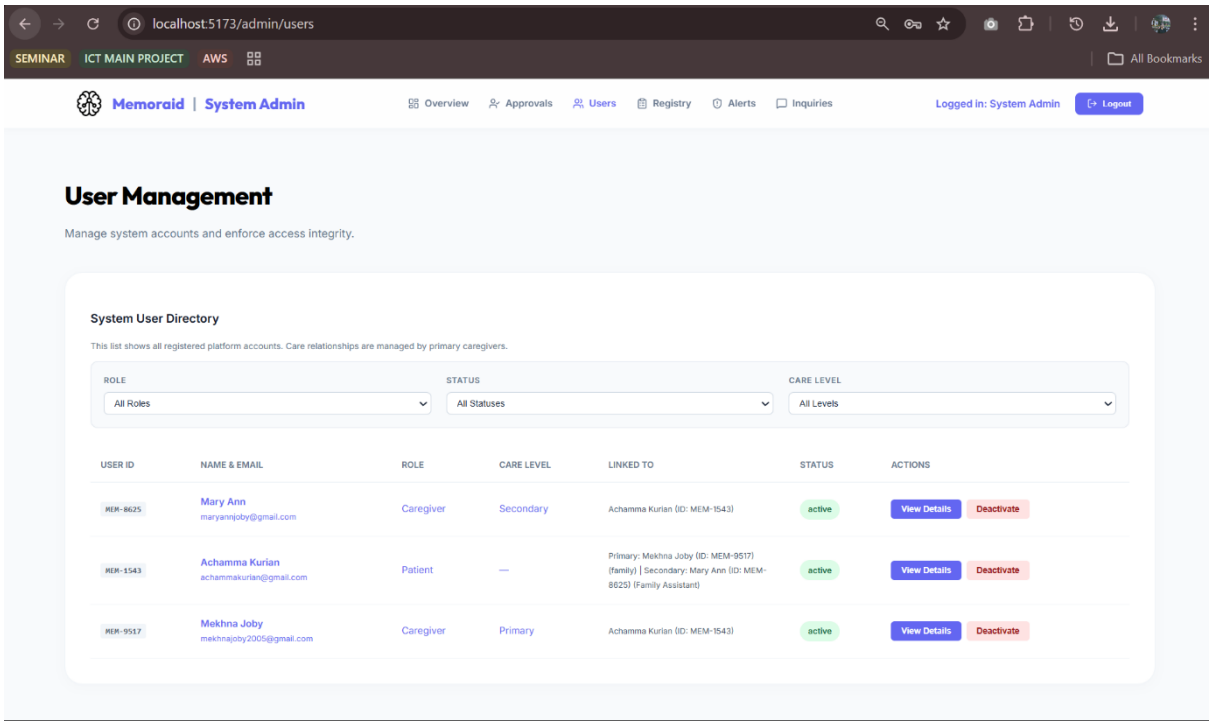
Dashboard overview



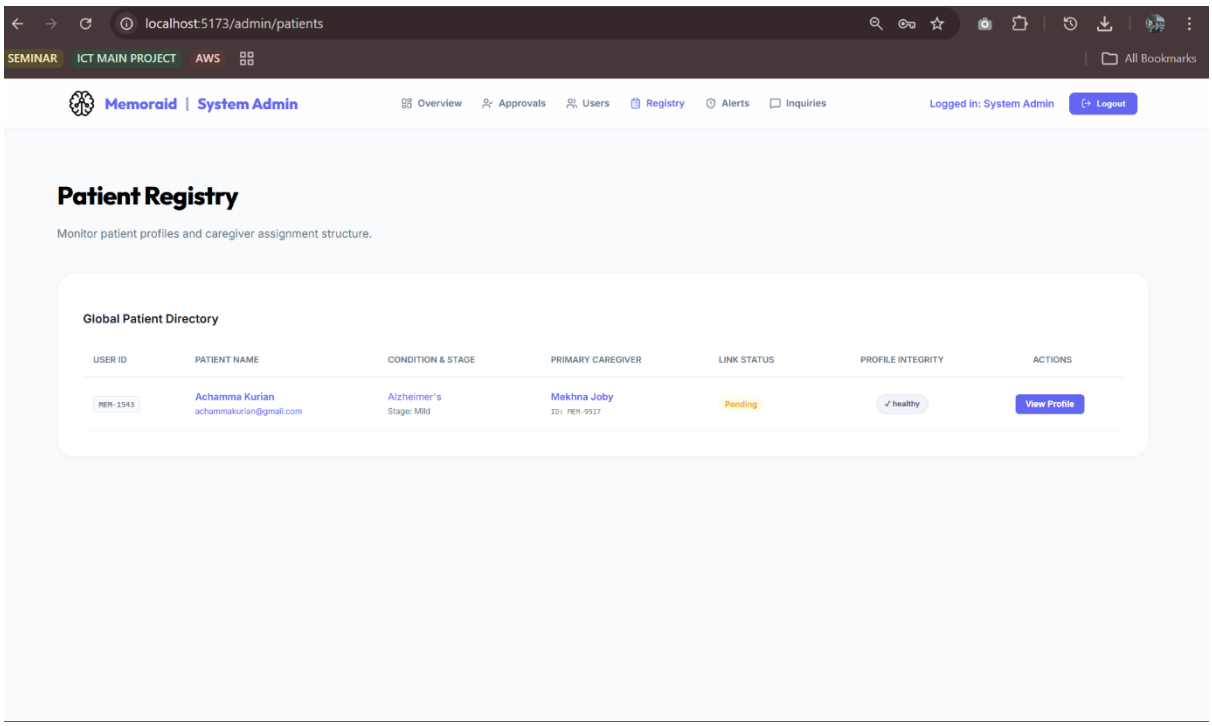
Approvals Page

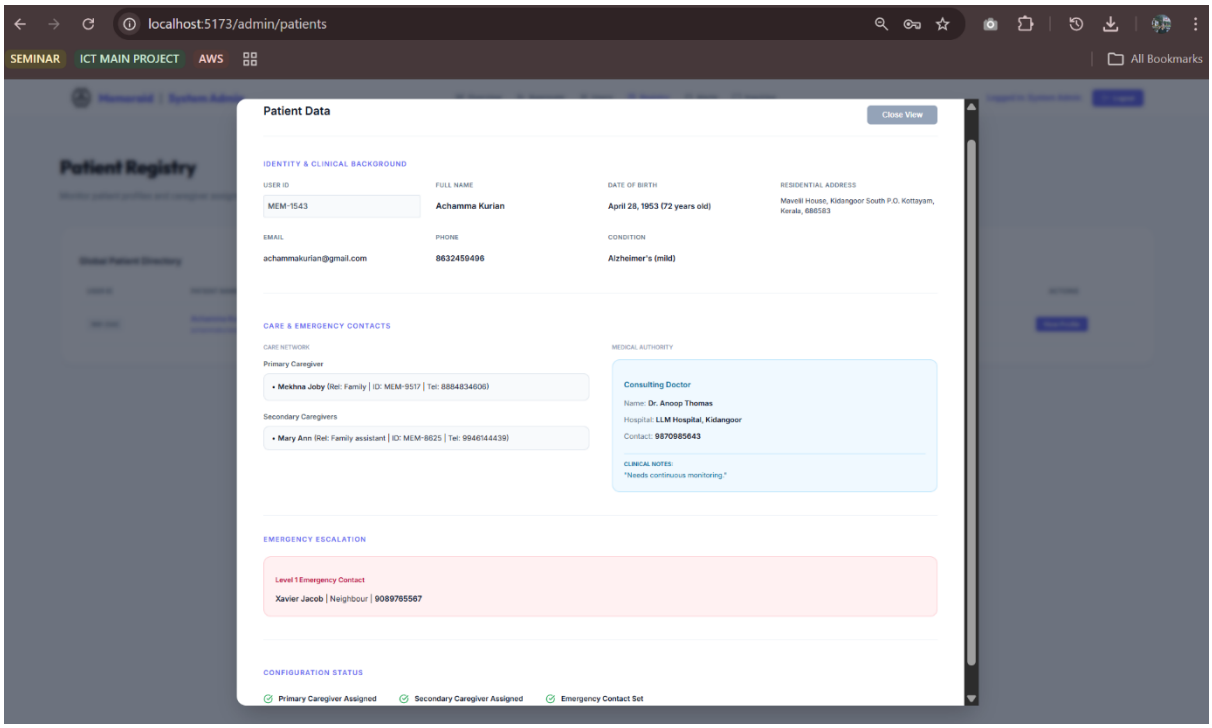


User Management

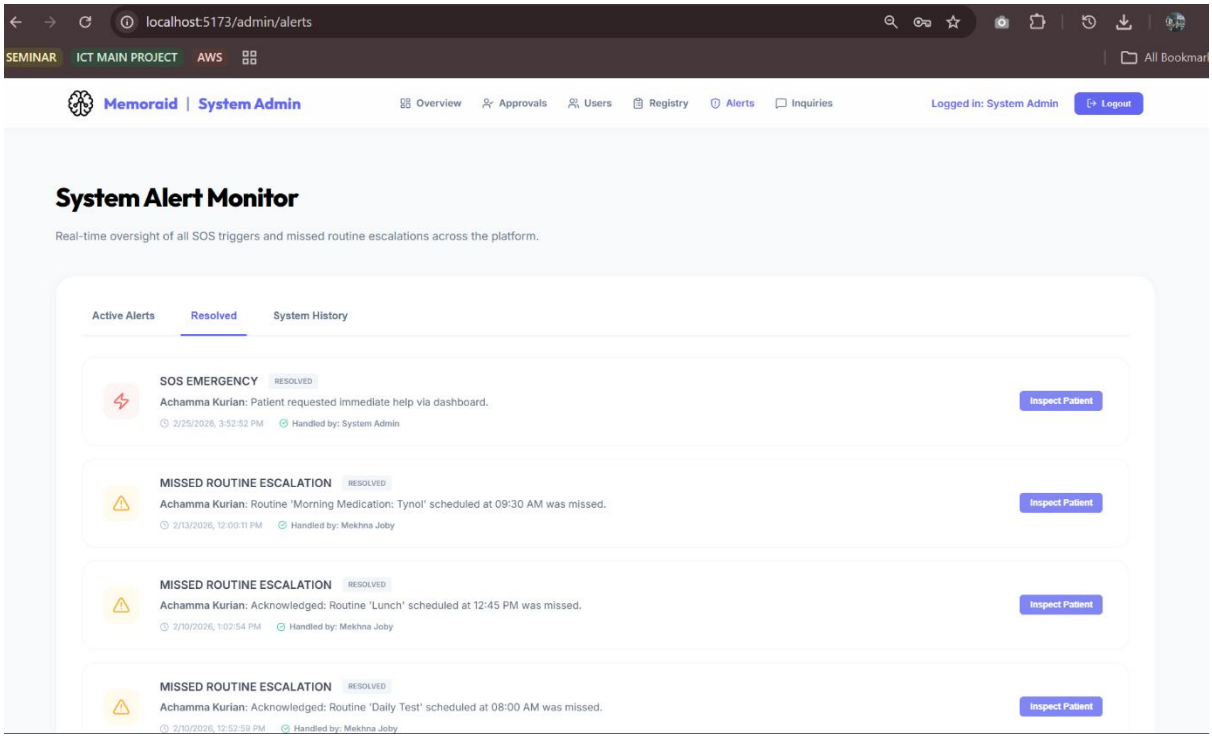


Patient Registry

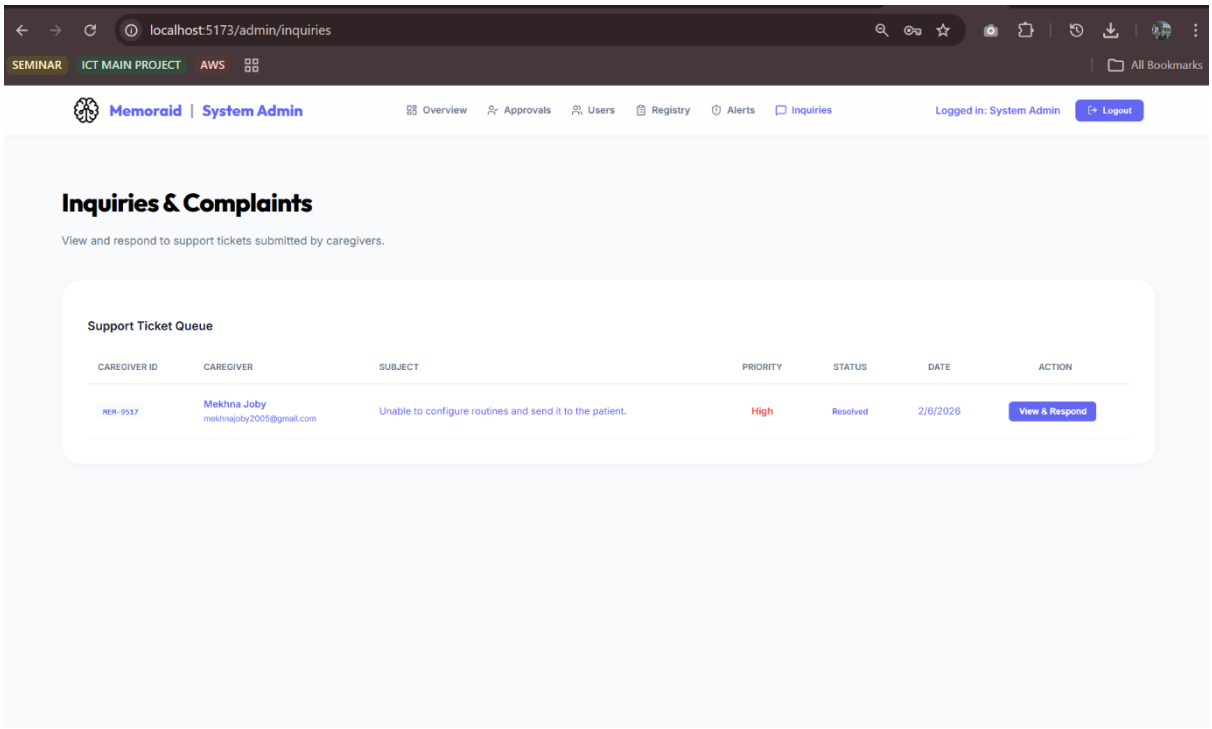



localhost:5173/admin/patients
SEMINAR
ICT MAIN PROJECT
AWS
All Bookmarks
Patient Registry
Patient Data
Close View
IDENTITY & CLINICAL BACKGROUND
USER ID
MEM-1543
FULL NAME
Achamma Kurian
DATE OF BIRTH
April 28, 1953 (72 years old)
RESIDENTIAL ADDRESS
Mavelil House, Kidangoor South P.O. Kottayam, Kerala, 686583
EMAIL
achammakurian@gmail.com
PHONE
8632459496
CONDITION
Alzheimer's (mild)
CARE & EMERGENCY CONTACTS
CARE NETWORK
Primary Caregiver
• Mekhna Joby (Rel: Family | ID: MEM-9517 | Tel: 8884834606)
Secondary Caregivers
• Mary Ann (Rel: Family assistant | ID: MEM-8625 | Tel: 9946144439)
MEDICAL AUTHORITY
Consulting Doctor
Name: Dr. Anoop Thomas
Hospital: LLM Hospital, Kidangoor
Contact: 9870985643
CLINICAL NOTES:
"Needs continuous monitoring."
EMERGENCY ESCALATION
Level 1 Emergency Contact
Xavier Jacob | Neighbour | 9089765567
CONFIGURATION STATUS
Primary Caregiver Assigned
Secondary Caregiver Assigned
Emergency Contact Set

Alerts Page

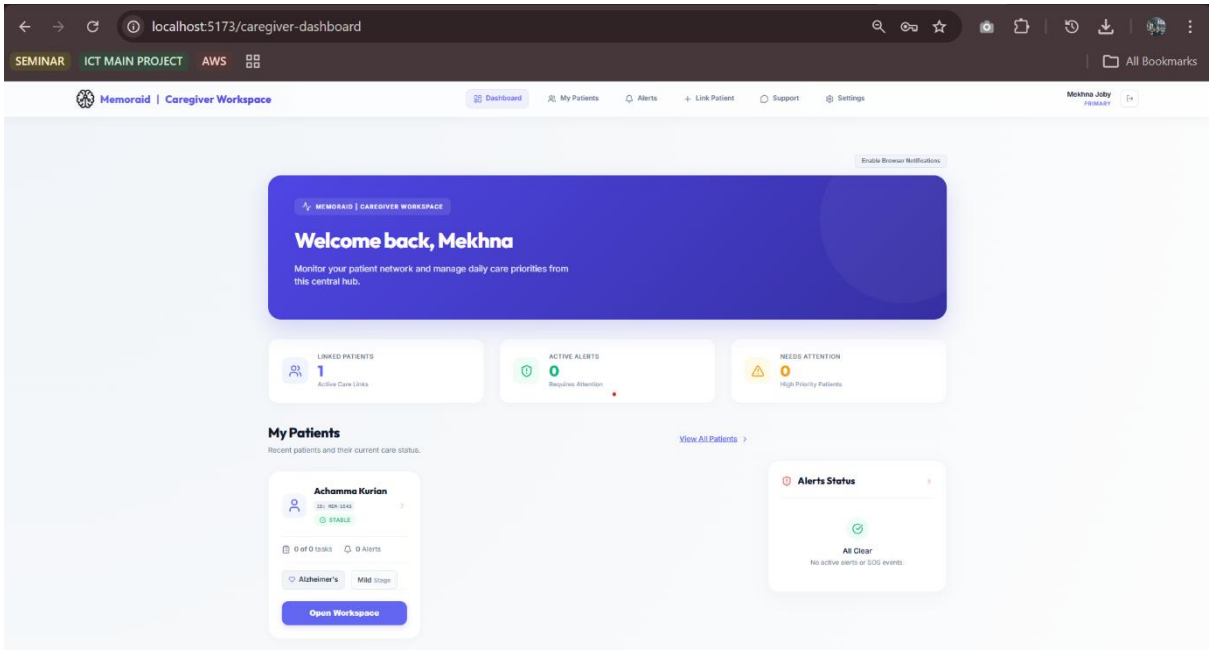

localhost:5173/admin/alerts
SEMINAR
ICT MAIN PROJECT
AWS
Memoraid | System Admin
Overview
Approvals
Users
Registry
Alerts
Inquiries
Logged in: System Admin
Logout
System Alert Monitor
Real-time oversight of all SOS triggers and missed routine escalations across the platform.
Active Alerts
Resolved
System History
SOS EMERGENCY
RESOLVED
Achamma Kurian: Patient requested immediate help via dashboard.
2/25/2026, 3:52:52 PM
Handled by: System Admin
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Routine 'Morning Medication: Tynol' scheduled at 09:30 AM was missed.
2/13/2026, 12:00:11 PM
Handled by: Mekhna Joby
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Acknowledged: Routine 'Lunch' scheduled at 12:45 PM was missed.
2/10/2026, 1:02:54 PM
Handled by: Mekhna Joby
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Acknowledged: Routine 'Daily Test' scheduled at 08:00 AM was missed.
2/10/2026, 12:52:59 PM
Handled by: Mekhna Joby
Inspect Patient

Inquiries and Complaints Page

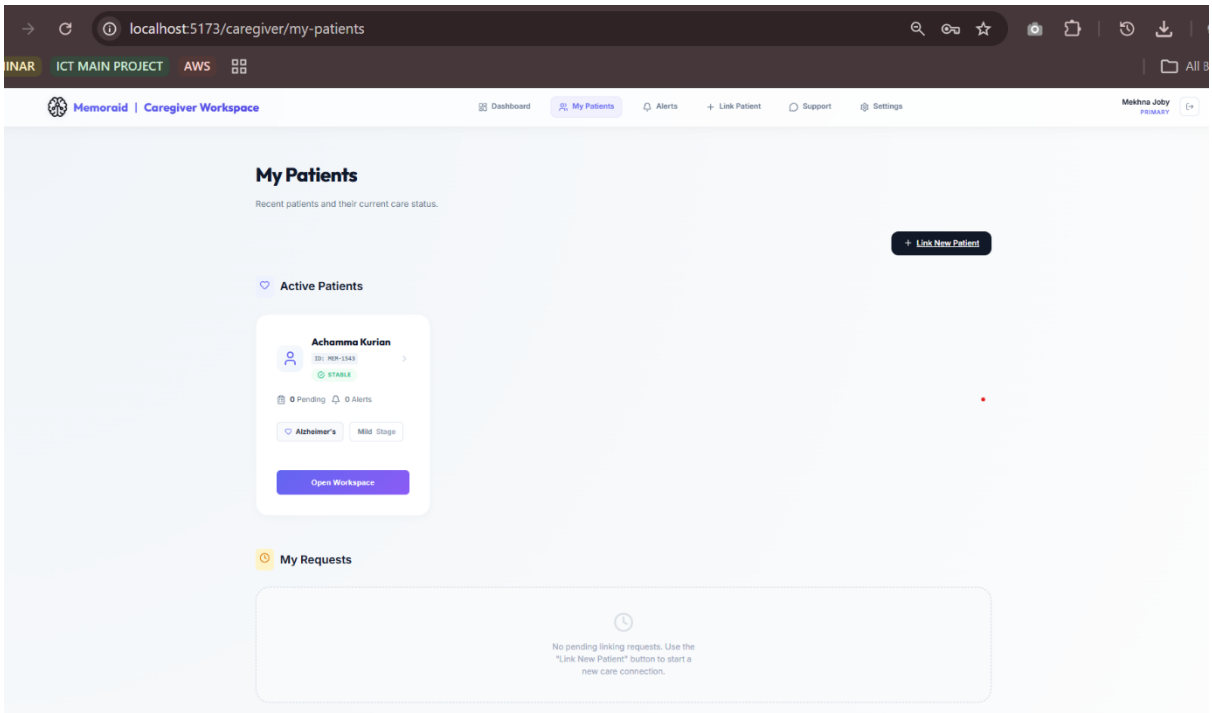


Caregiver Module

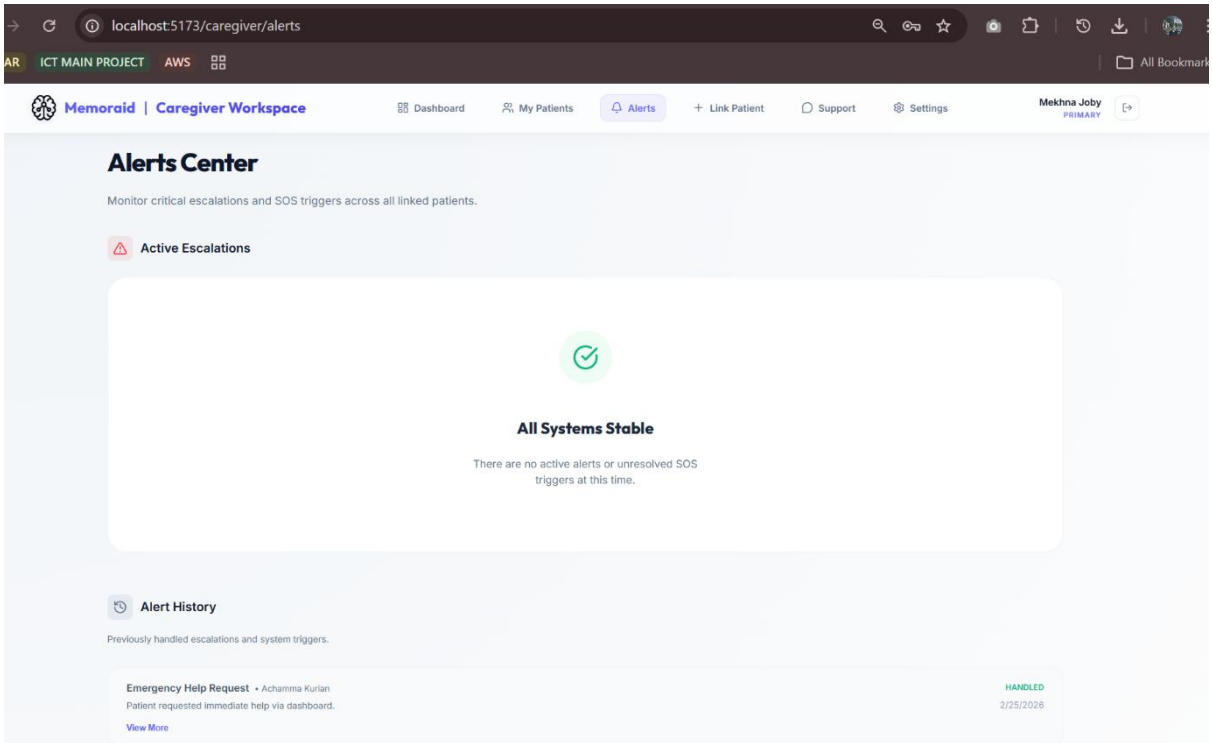
Dashboard Overview



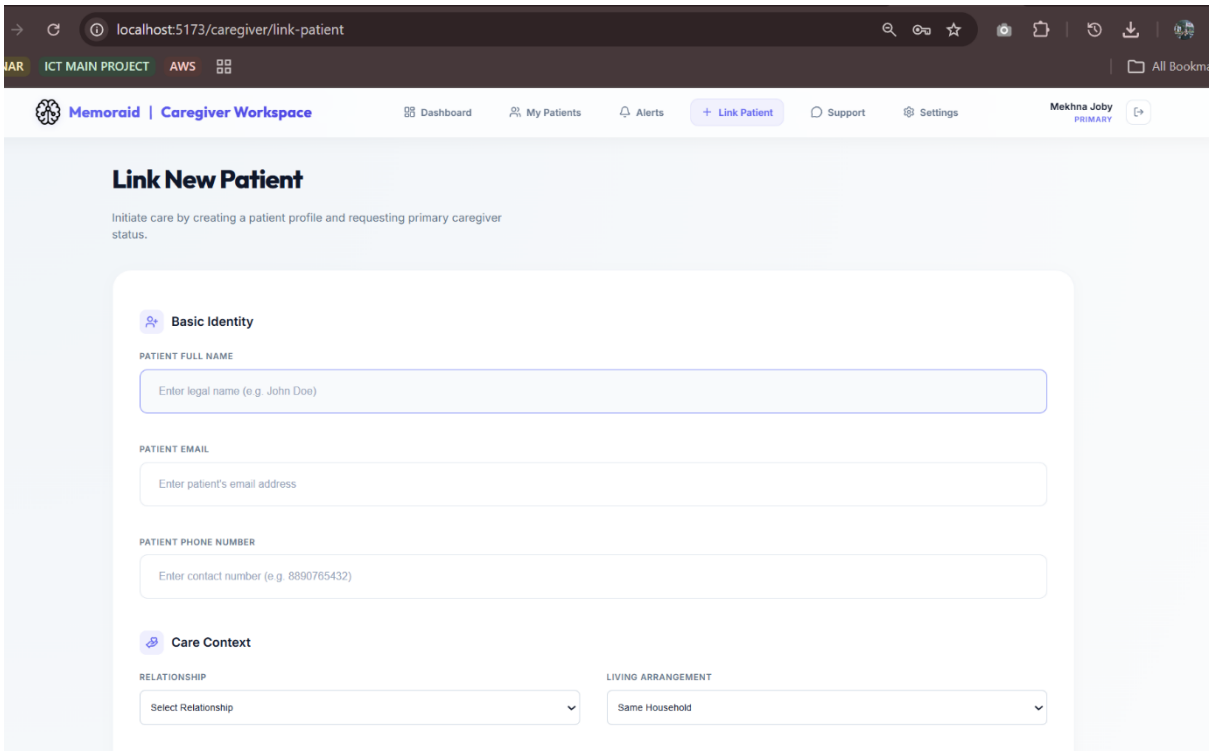
My Patients Page



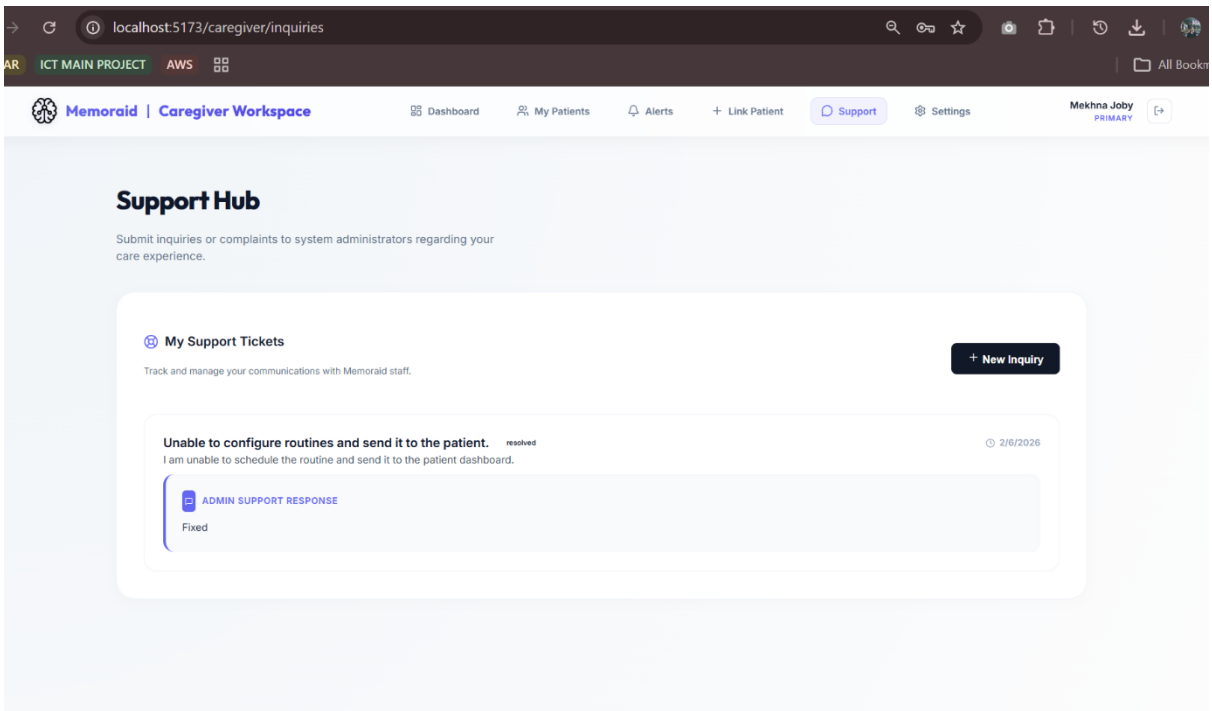
Alerts Page



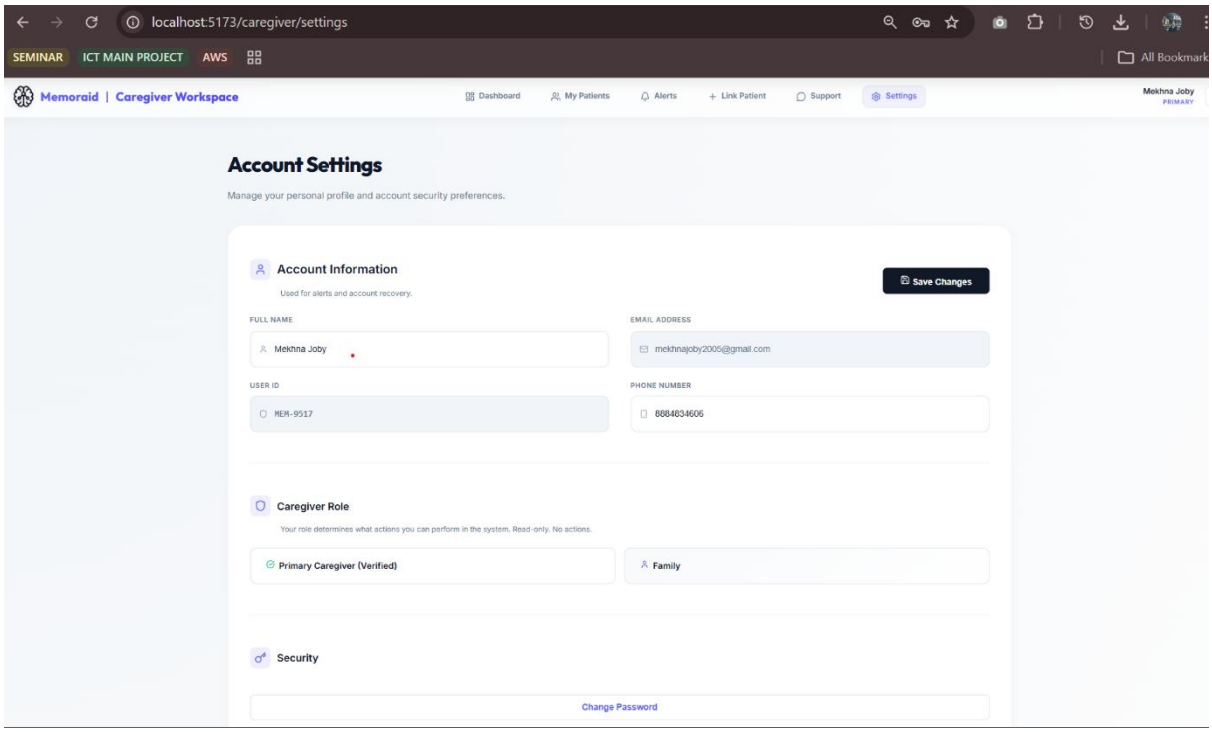
Link Patient Page



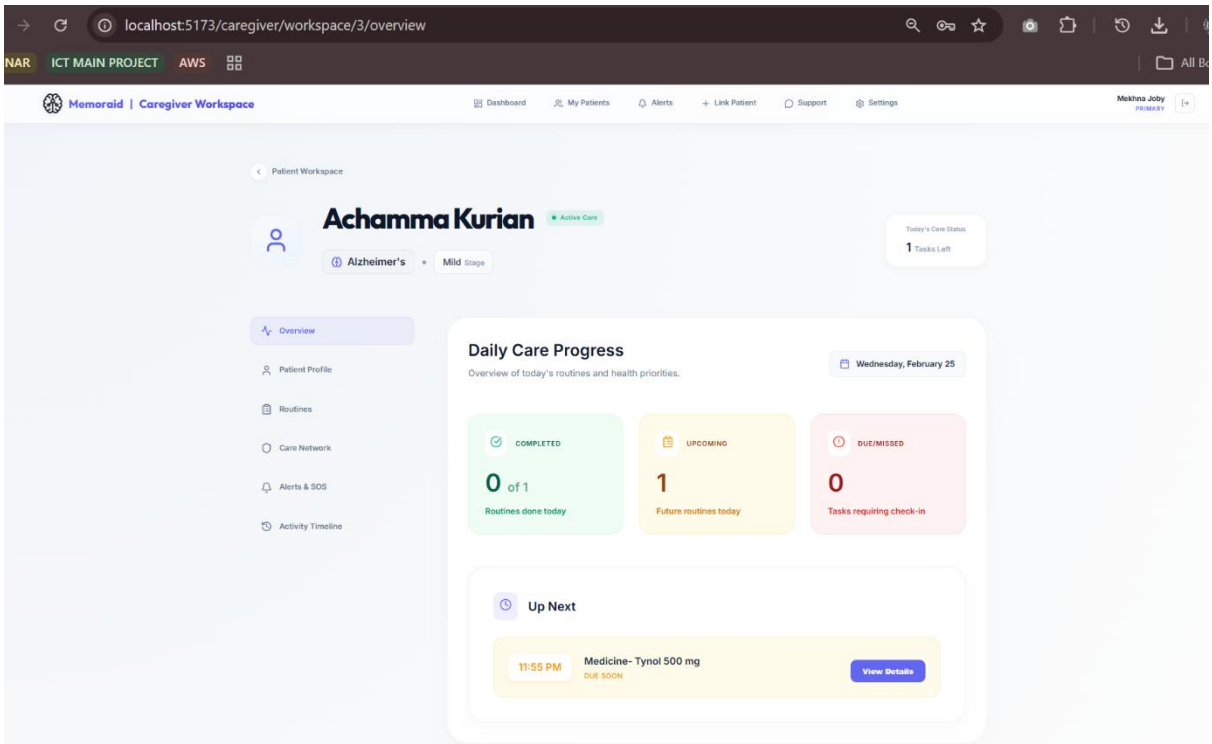
Support Center



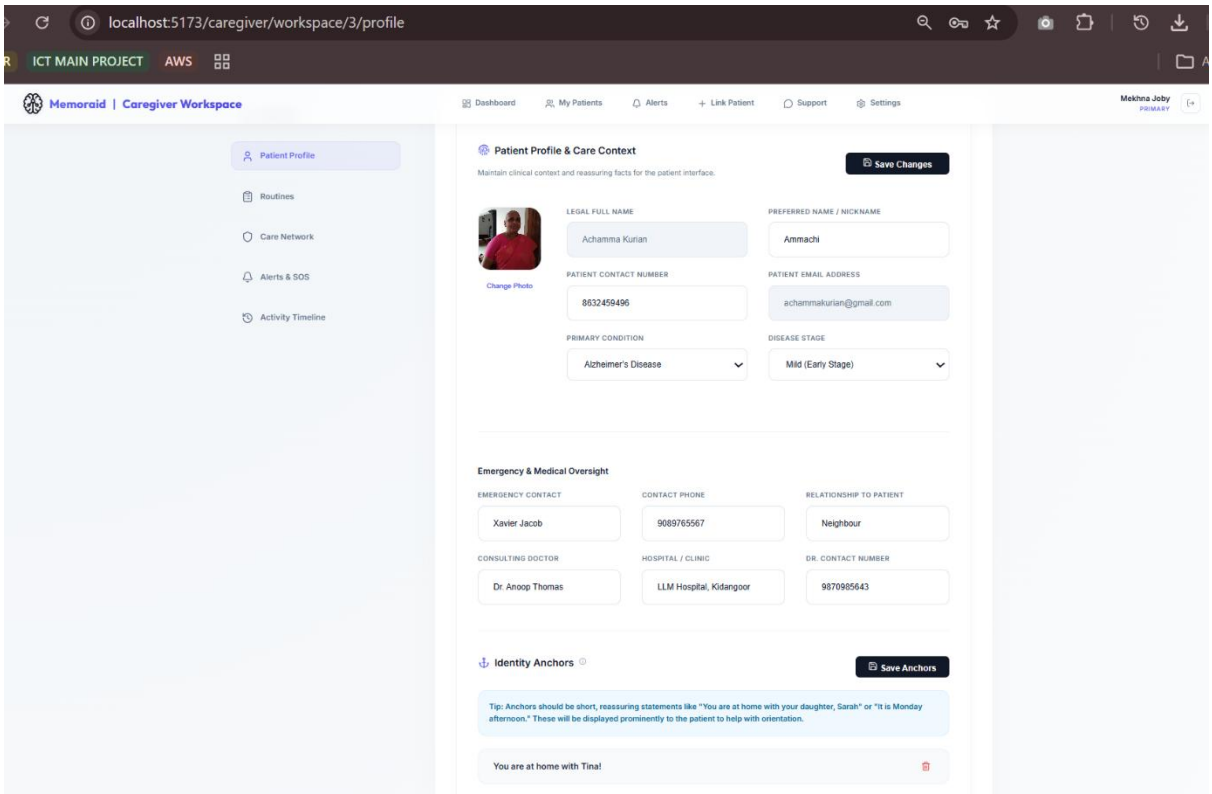
Settings Page



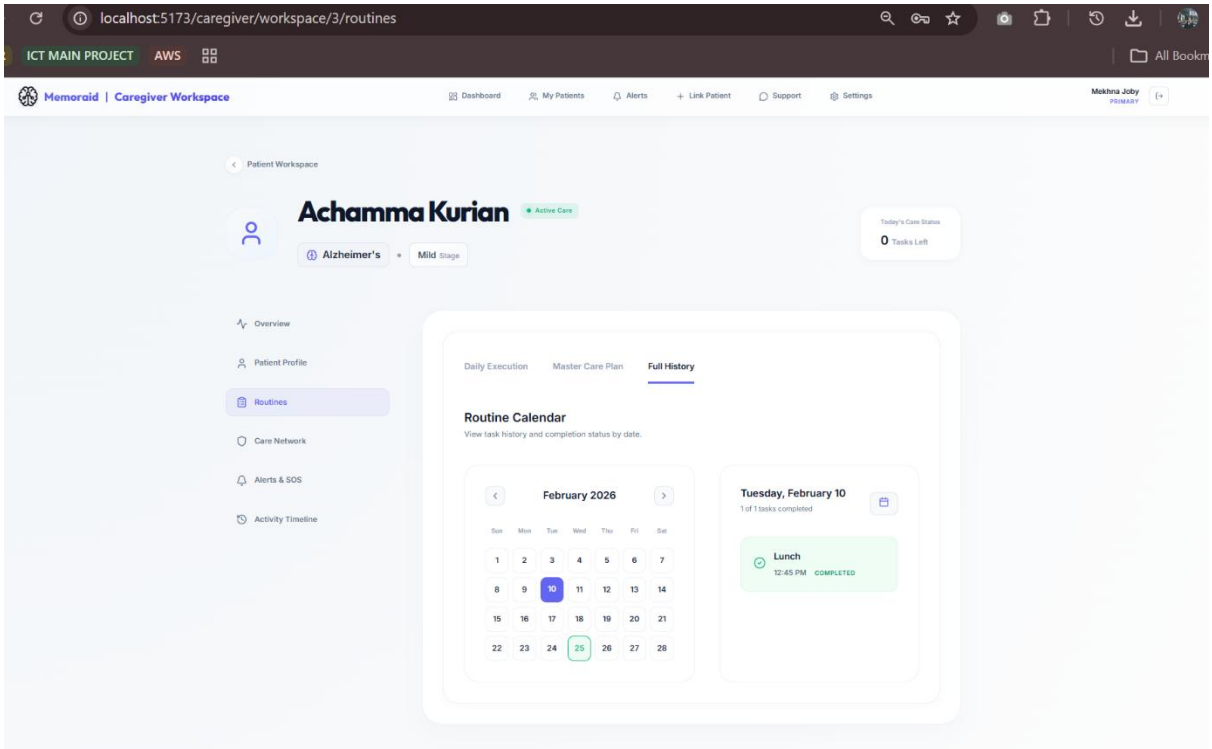
Patient Workspace Overview

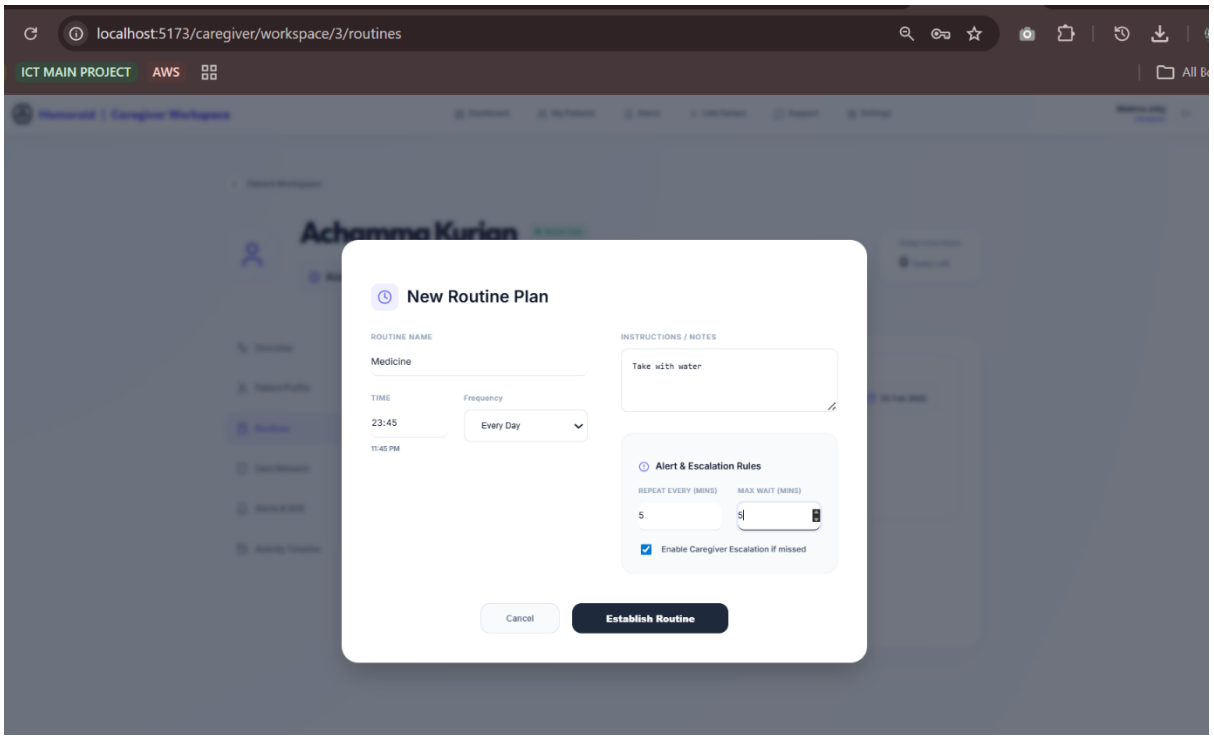


Patient’s Profile Page



Routines Page

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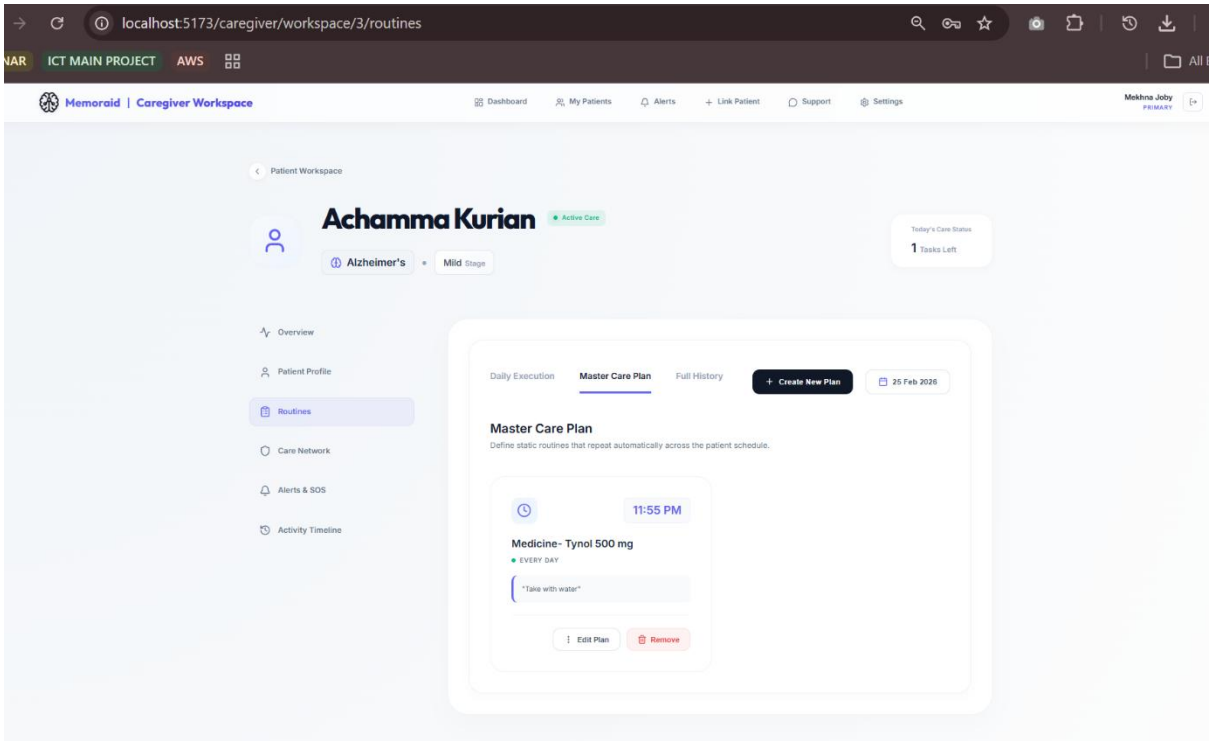
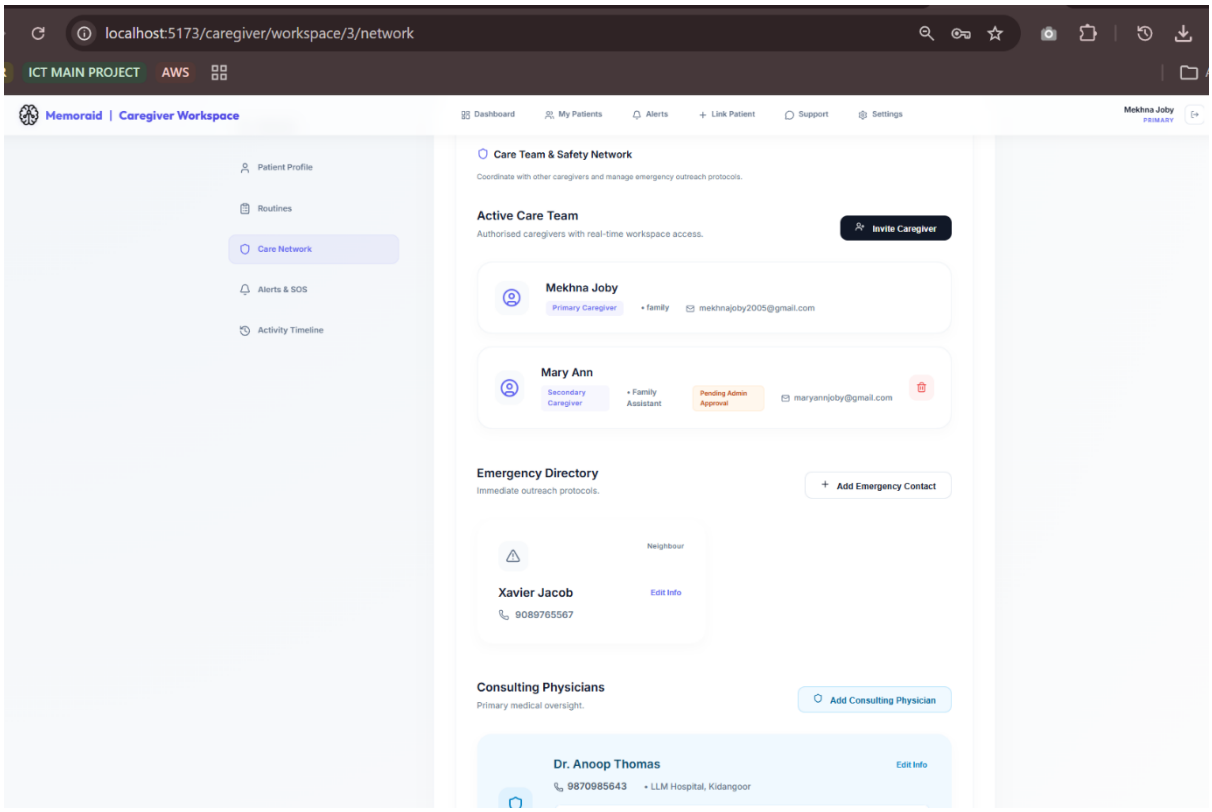
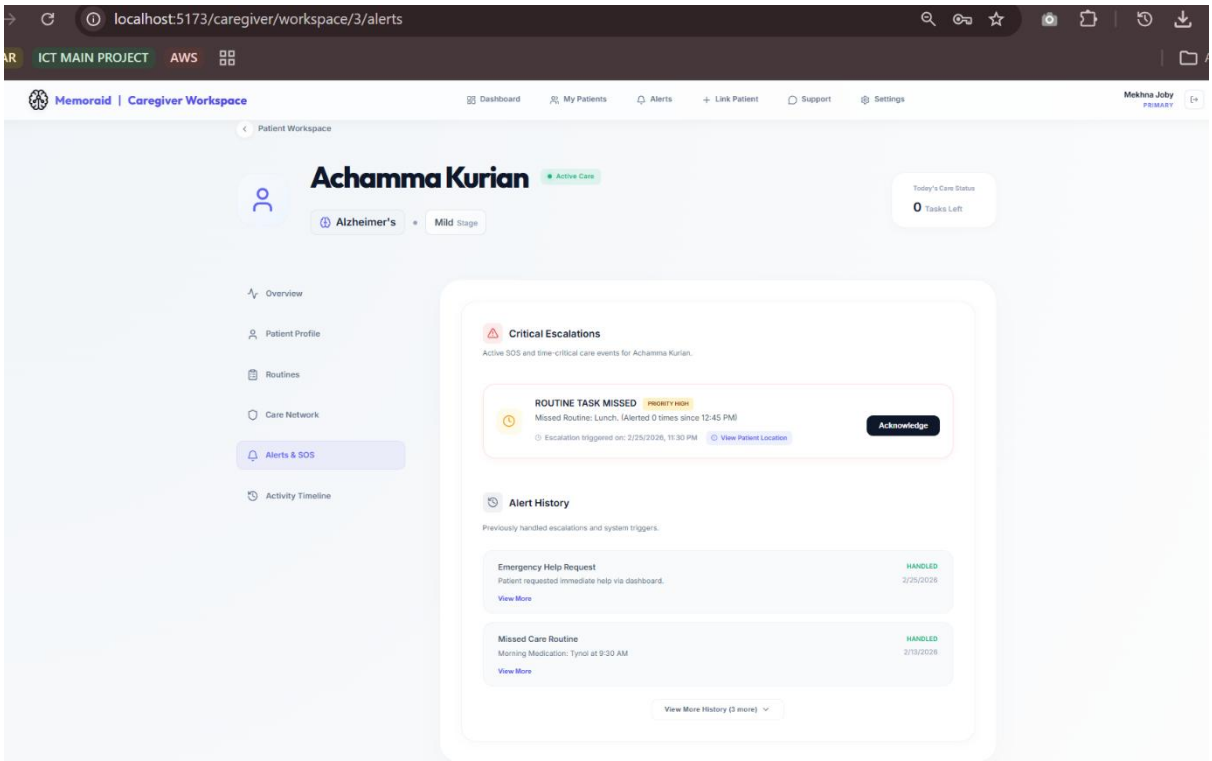
ICT MAIN PROJECT
AWS
New Routine Plan
ROUTINE NAME
Medicine
INSTRUCTIONS / NOTES
Take with water
TIME
23:45
11:45 PM
Frequency
Every Day
Alert & Escalation Rules
REPEAT EVERY (MINS)
5
MAX WAIT (MINS)
5
Enable Caregiver Escalation if missed
Cancel
Establish Routine


localhost:5173/caregiver/workspace/3/routines
ICT MAIN PROJECT
AWS
Memoraid | Caregiver Workspace
Dashboard
My Patients
Alerts
Link Patient
Support
Settings
Mekhna Joby
PRIMARY
Patient Workspace
Achamma Kurian
Active Care
Alzheimer's
Mild Stage
Today's Care Status
1 Tasks Left
Overview
Patient Profile
Routines
Care Network
Alerts & SOS
Activity Timeline
Daily Execution
Master Care Plan
Full History
+ Create New Plan
25 Feb 2026
Master Care Plan
Define static routines that repeat automatically across the patient schedule.
11:55 PM
Medicine- Tynol 500 mg
EVERY DAY
"Take with water"
Edit Plan
Remove

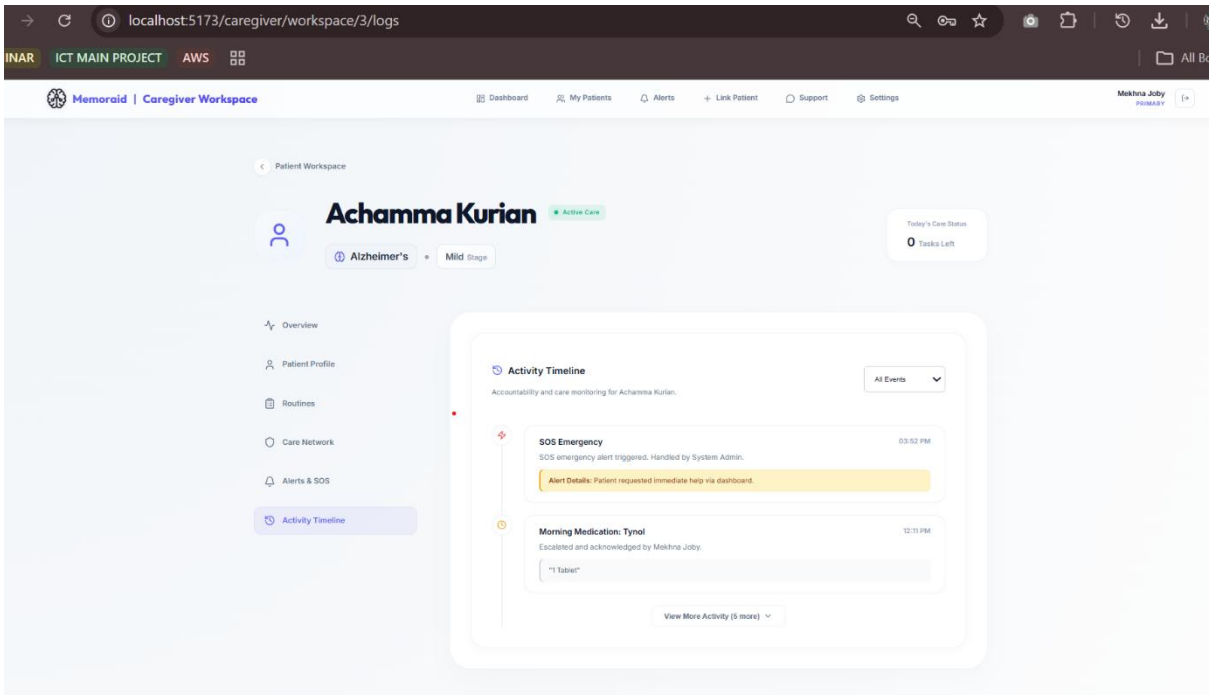
Care Network Page



Alerts & SOS Page

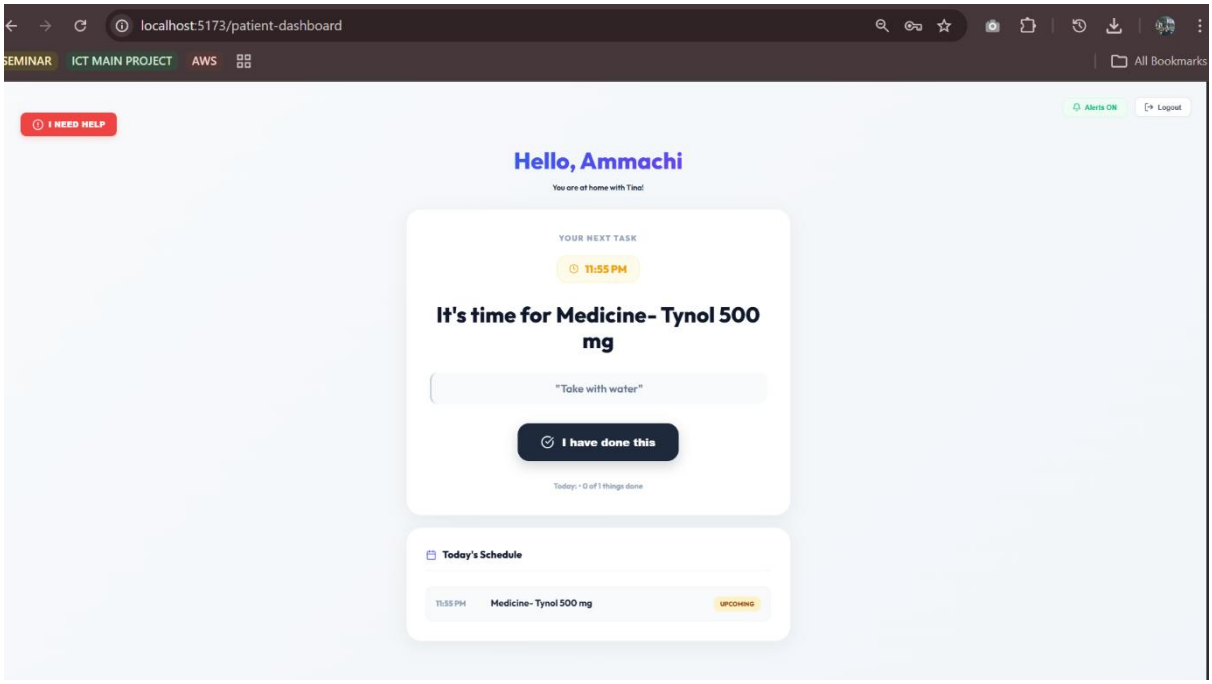
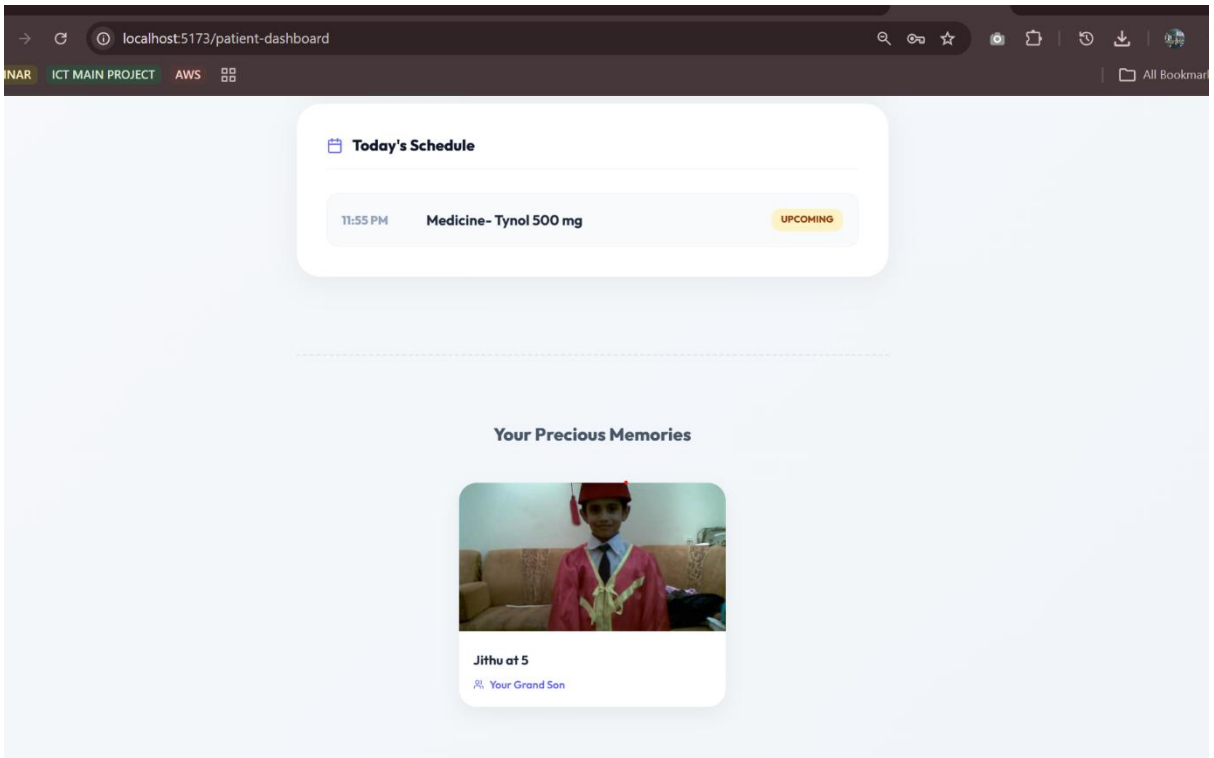


Activity Timeline Page



Patient Module

Dashboard Overview



Memory Gallery